

SignalShap: Exact Ranking-Stage Attribution of Sources in Hybrid Recommenders

Mouad Louhichi [ORCID](#)^{1*}, Redwane Nesmaoui¹ and Mohamed Lazaar¹

^{1*}National Higher School of Computer Science and Systems Analysis (ENSIAS), Mohammed V University in Rabat, Rabat, Morocco.

*Corresponding author(s). E-mail(s): mouad_louhichi@um5.ac.ma;
Contributing authors: redwane.nesmaoui@um5.ac.ma;
mohamed.lazaar@ensias.um5.ac.ma;

Abstract

Source ablation and Shapley attribution are often treated as competing estimates of one quantity: how much a component of a hybrid recommender is worth. They answer different questions, and conflating them gives the wrong instrument for the decision at hand.

We present *SignalShap*, which recasts source attribution as a five-player cooperative game — collaborative filtering, content, popularity, recency and sequential order — whose payoff is a fixed-candidate normalised discounted cumulative gain at rank 10 (NDCG@10) on a coalition-independent candidate set. With five players the Shapley value follows from exact enumeration of all $2^5 = 32$ coalitions, so the aggregation carries no sampling error. Six analytic games with known Shapley vectors reproduce exactly and efficiency holds to 6.9×10^{-18} .

Across three timestamped corpora spanning a $36\times$ density range, two of three systems contain a source on which the two rules disagree in *sign* by a material margin, at every regularisation strength we tested. Yet in an end-to-end retirement simulation, leave-one-out predicts the observed cost of removal *better* than Shapley on all three corpora and over ten seeds (mean rank agreement **0.94–1.00** against **0.20–0.80**), identifying the cheapest source to retire on **80–100%** of seeds against **0–30%**. SignalShap allocates the value of a declared ranking-stage game; leave-one-out estimates the effect of removal. We therefore claim credit allocation, not retirement guidance, and identify candidate recall as an applicability test that only one corpus passes. Code and artefacts are released.

Keywords: Shapley value, cooperative game theory, hybrid recommender systems, ranking-stage attribution, model-component valuation, performance attribution

1 Introduction

A production hybrid recommender is an ensemble of architecturally distinct scorers. A collaborative-filtering component exploits co-occurrence, a content component exploits item metadata, a popularity component exploits global frequency, a recency component exploits temporal locality, and a sequential component exploits order. Engineering decisions — which component to invest in, which to retire — rest on estimates of how much each contributes. We show in Section 13 that these are two distinct questions requiring different instruments, and that this paper addresses the first. In practice those estimates come from *leave-one-out* (LOO) *ablation*: disable one component, measure the drop in ranking quality, and attribute that drop to the component.

Ablation and Shapley allocation estimate *different quantities*, and the distinction is the subject of this paper. Leave-one-out measures the conditional effect of removing a component from the *complete* deployed system while every other component remains. That is a valid estimand, and dependence between components does not invalidate it — it is exactly the right quantity for a removal decision, as our own experiments in Section 13 confirm.

What it is not is a complete rule for dividing credit. Real hybrids overlap by construction: matrix factorisation on implicit signals is strongly popularity-driven, so *cf* and *pop* carry shared information. When two components are mutually substitutable, removing either alone changes little and both receive near-zero conditional value, even though the capability they share is doing work. A rule that allocates the system’s value among interacting components must average over the situations in which each might be needed, including those where no substitute is available. Neither estimand dominates; they answer different questions, and this paper measures where they diverge.

Cooperative game theory answers exactly that question. The Shapley value [33] is the unique credit assignment satisfying efficiency, symmetry, null-player, and additivity, and it evaluates each player against *every* coalition rather than only against the grand coalition. Its adoption in machine learning has been broad [6, 25], but almost entirely at the granularity of *input features*. Our object of study is different: the players are *architectural sources*, and the payoff is a ranking metric.

Figure 1 summarises the pipeline. That change of granularity buys something unusual. Feature-level Shapley must approximate — the coalition lattice over hundreds of features is intractable, so KernelSHAP and its relatives sample and inherit estimator variance. With five architectural sources the lattice has $2^5 = 32$ elements and the exact value is a short sum over $2^5 = 32$ terms, which we enumerate exactly. Second, and more importantly, *verifiability*: a small player set makes it possible to construct cases in which the attribution must satisfy a known constraint, and check that the method does. Section 9 does this for symmetry and efficiency on real fitted games.

1.1 Contributions

- **C1.** A precisely declared ranking-stage cooperative game over heterogeneous recommender sources, with a coalition-independent candidate set that makes $v(S)$ comparable across coalitions, a per-user formulation, and exact enumeration for small source sets (Section 4).

- **C2.** A formal characterisation of source allocation under redundancy: efficiency and per-user decomposition, a counterexample showing the redundancy–positivity property is **false** without monotonicity, a corrected two-sided ε -substitutability result, and the observation that Shapley credit is *not* conserved under replication (Section 6).
- **C3.** A validation protocol for source attribution combining analytic games with known Shapley vectors, controlled duplicate injection, and efficiency, null-player, and additivity diagnostics (Sections 9 and 10). These establish that the aggregation is implemented correctly — a precondition for interpreting anything computed with it, and one the field rarely checks.
- **C4 (primary).** An empirical comparison of three estimands — refitted-head ranking-stage, fixed-head, and end-to-end retrieval — showing that Shapley allocation and leave-one-out removal cost answer *different questions*. Over ten seeds on all three corpora, leave-one-out predicts retirement cost better than Shapley (mean rank agreement 0.94–1.00 against 0.20–0.80) and identifies the cheapest source to retire on 80–100% of seeds against 0–30%. We pair this with candidate recall as a necessary applicability test, and report where our own method should not be used (Sections 11 and 13).

What this paper does not claim

We do not claim that attribution-derived fusion weights improve ranking quality: they are practically equivalent to a globally tuned head under a pre-specified margin, and we report that negative result in Appendix A rather than as a contribution. We make no state-of-the-art claim against LightGCN or SASRec; those models appear as contextual references only, and received smaller tuning budgets than a performance comparison would require. And we do not claim that the sign of an attribution prescribes retaining or removing a source — Section 13 shows it does not.

1.2 Research questions

RQ1 Does a fixed-candidate, coalition-independent cooperative game over architectural sources admit an exact additive decomposition of ranking quality, and does the implementation satisfy the Shapley axioms on real fitted games?

RQ2 Does source-level Shapley attribution differ materially from leave-one-out ablation, and is the difference associated with measured redundancy between sources?

RQ3 Do the ranking-stage, fixed-head, and end-to-end estimands agree, and which of them predicts the measured cost of retiring a source?

What is not in this paper

Graph-neural or Transformer players; causal attribution; online or counterfactual evaluation. All attribution is *within* the declared game and the fixed candidate set.

Roadmap

Section 2 positions the work against component-level and ranking-specific attribution. Section 3 collects the cooperative-game definitions we rely on. Section 4 defines the

game, Section 5 gives the algorithm and its cost, and Section 6 the formal properties, including a counterexample showing that redundancy alone does not guarantee positive credit. Sections 7–8 describe the corpora and the two preconditions that gate everything after them. Sections 9–10 validate the implementation against constraints known in advance. Section 11 reports the attributions, Section 13 compares estimands and reports the retirement result, and Sections 14–15 discuss scope and threats. Appendix A reports a negative result.

2 Related Work

Shapley values in machine learning

SHAP [25] unified additive feature attribution; extensions cover trees [26], data valuation [8], and federated contribution [38]. Surveys [6, 31] note that essentially all practical estimators sample, because the coalition lattice is exponential in the number of players. Our setting sidesteps this: five players, 32 coalitions, exact enumeration.

Feature-level XAI versus source-level attribution

SHAP, LIME [29], and Integrated Gradients [35] answer “which input features moved this prediction?”. We answer “which architectural component earned the system’s ranking quality?”. The unit of attribution differs, the payoff is a ranking metric rather than a scalar output, and the actionable decision differs — ours is the allocation of system value across subsystems.

Attribution to model components and to performance

Closer to our unit of analysis than feature-level XAI is work attributing behaviour to *model components* rather than inputs. Neuron Shapley [9] treats individual neurons as players with a performance payoff, and ensemble games [30] treat whole classifiers as players and allocate ensemble accuracy among them — the closest structural precedent for what we do, since both take trained models as players and a performance metric as the characteristic function. Related work attributes changes in *performance* rather than individual predictions [32, 41]. Our framing of Shapley as “almost entirely feature-level” would be wrong; the correct statement is that component-level valuation exists and that this paper extends it to architectural sources in a ranking system.

Shapley beyond supervised prediction

Shapley attribution has also been applied to unsupervised settings, where the absence of labels removes the natural payoff that classification or regression supplies. Our own earlier work [24] attributes cluster structure to input features by pairing approximated Shapley values with multi-level clustering. That work shares this paper’s premise — that a cooperative game is a defensible way to divide credit in a model whose internals resist inspection — but differs on every axis that matters here: the players are input features rather than trained source models, the payoff is cluster quality rather than ranking utility, and the values are approximated rather than enumerated exactly. The present paper moves the unit of analysis from features to architectural components, which is what makes exact enumeration affordable and what makes the retirement

question meaningful in the first place: one can switch off a recommender source, but not a feature of a clustering.

Shapley for ranking

A recent line develops Shapley attribution specifically for ranked outputs. RankSHAP [5] axiomatises feature attribution for learning-to-rank [1] and evaluates against ranking metrics; RankingSHAP [15] gives a listwise formulation; ShaRP [28] defines games for score, rank, top- k , and pairwise-preference outcomes. Within recommendation, triplet importance [13] values training triplets by ranking utility. All of these take *features* or *training data* as players and approximate the value by sampling. SignalShap differs on both axes: the players are architectural source models, and with five players the value is computed by exhaustive enumeration rather than estimated. The distinguishing design choice is the coalition-independent candidate set with a coalition-refitted fusion head, which makes coalition values comparable across a subset lattice in a two-stage retrieval-then-rank architecture — a construction these works do not require because their players do not affect retrieval.

Which Shapley, and which game

Shapley uniqueness holds *once a game is fixed*; it does not select the game. Sundararajan and Najmi [34] show that plausible operationalisations differ materially, which is why we state our candidate construction, coalition head, and baseline explicitly and treat them as declared modelling choices rather than as the canonical formulation. Alternative cooperative values — Banzhaf and other semivalues [22] — and interaction-aware extensions such as the Shapley interaction index [10] and the Shapley–Taylor index [36] are natural comparators, as are faithful [37] and regression-based [7] interaction formulations. Because every game we fit is non-monotone, recent work on negative interactions in non-convex games [3] is also directly relevant. Because redundancy between sources is central to our setting we evaluate Banzhaf, a uniform semivalue, and the Grabisch–Roubens interaction index rather than deferring them (Section 10.2); we do not compute Shapley–Taylor, and Section 10.2 explains why the distinction matters. Replication sensitivity of Shapley allocations [11] is directly relevant to our duplicate-injection design and is discussed in Section 9.

Hybrid recommenders and ablation practice

Burke’s taxonomy [2] classifies hybridisation strategies; weighted and feature-combination hybrids dominate practice. Evaluation of component worth is predominantly by ablation [40]. Recent recommender XAI work [42] explains *recommendations*, not *architectures*.

3 Background

We collect the definitions the rest of the paper assumes. Readers familiar with cooperative games may skip to Section 4.

Cooperative games and the Shapley value

A transferable-utility game is a pair $(\mathcal{G}, v)$ with a finite player set $\mathcal{G}$, $n = |\mathcal{G}|$, and a characteristic function $v : 2^{\mathcal{G}} \rightarrow \mathbb{R}$ giving the value each coalition achieves on its own. The Shapley value [33] allocates $v(\mathcal{G})$ among the players by averaging each player’s marginal contribution over all orderings, and is the unique allocation satisfying efficiency, symmetry, null player, and additivity. Nothing in the axioms selects v : uniqueness is conditional on a game the modeller declares, which is why Section 4 states ours explicitly and Section 13 compares alternatives.

Semivalues

Relaxing efficiency yields a family of *semivalues*, each averaging marginal contributions under a different weighting of coalition sizes. The Banzhaf value weights all 2^{n-1} coalitions equally; a uniform semivalue weights each coalition *size* equally. Neither sums to $v(\mathcal{G})$, so only the induced ordering is comparable across values. We report both as robustness checks in Section 10.2.

Interaction indices

Pairwise indices measure whether two players are complementary or redundant beyond their main effects, via the second-order discrete derivative $v(S \cup \{a, b\}) - v(S \cup \{a\}) - v(S \cup \{b\}) + v(S)$. They differ in how these terms are weighted: the Shapley interaction index of Grabisch and Roubens [10] and the Shapley–Taylor index [36] use different coefficients and are *not* interchangeable. We use the former and say so, having previously conflated the two.

Leave-one-out, in two distinct forms

Leave-one-out ablation reports the single marginal contribution at the grand coalition, $v(\mathcal{G}) - v(\mathcal{G} \setminus \{g\})$. It is a valid estimand for the effect of removing g while everything else remains, and it is not an approximation to the Shapley value; it is one of the 2^{n-1} terms the Shapley value averages.

Which v it is computed on matters, and conflating the two is easy. Write

$$\text{LOO}_{\text{rank}}(g) = v_{\text{rank}}(\mathcal{G}) - v_{\text{rank}}(\mathcal{G} \setminus \{g\}), \quad \text{LOO}_{\text{e2e}}(g) = v_{\text{e2e}}(\mathcal{G}) - v_{\text{e2e}}(\mathcal{G} \setminus \{g\}). \quad (1)$$

LOO_{rank} removes g from the *fusion head only*: the candidate set is still the one all five sources retrieved, so g keeps its retrieval role. LOO_{e2e} removes g from retrieval as well, rebuilding C_u from the survivors. The second is what actually happens when a source is switched off in production, and so LOO_{e2e} *equals* the observed single-source retirement loss by construction — it is a definition, not a finding. LOO_{rank} is a different, cheaper quantity that requires no re-retrieval. The empirical question of Section 13 is therefore whether the cheap ranking-stage quantity predicts the expensive end-to-end one, and whether it does so better than the ranking-stage Shapley value computed on the same game.

Ranking-stage versus end-to-end games

A production recommender retrieves candidates and then ranks them. A *ranking-stage* game holds retrieval fixed and varies only the sources available to the ranker, so coalition values are comparable. An *end-to-end* game lets each coalition retrieve its own candidates, which is closer to a deployment change but makes $v(S)$ depend on two mechanisms at once. We define the former and measure both (Section 13).

NDCG with one relevant item

Our protocol holds out a single interaction per user, so for a ranking π_u with the held-out item at position r , the normalised discounted cumulative gain [20] reduces to a single term: $\text{NDCG}@10(\pi_u; y_u) = 1/\log_2(r+1)$ if $r \leq 10$ and 0 otherwise, the ideal DCG being 1. The metric is therefore a step function of rank, which is why coalition values move in discrete jumps and why near-ties require an explicit tolerance.

4 The Cooperative Game

Let $\mathcal{G} = \{cf, ct, pop, rec, seq\}$ with $|\mathcal{G}| = 5$.

4.1 Coalition-independent candidates

For user u and source g , let $\text{top-}N_g^{(g)}(u)$ be that source’s own top- N_g items, and set

$$C_u = \text{Trunc}_{N_{\max}^{(d)}}\left(\bigcup_{g \in \mathcal{G}} \text{top-}N_g^{(g)}(u)\right), \quad |C_u| \leq N_{\max}^{(d)}. \quad (2)$$

Here d indexes the dataset, since the cap is frozen per corpus (Table 1), and Trunc_N retains the first N items after sorting the union by the key (best rank of the item across $g \in \mathcal{G}$, source order in $\mathcal{G}$, item index). All three components are seed-independent, so C_u is deterministic; the bound in (2) holds by construction rather than by assumption. Sources contribute equal depth, $N_g = N$ for all g , with N grown until the union reaches the cap.

The tie-break depends on source order, and we measured how much

Truncation breaks ties by best per-source rank, then source order, then item index. The middle key is arbitrary: relabelling the sources can change C_u . A reviewer asked whether this matters, so we measured it rather than argue. Across all 120 permutations of the five sources on MovieLens ($N_{\max} = 600$, 60 probe users), only one permutation reproduces the canonical candidate sets exactly, but the worst-case mean Jaccard overlap with them is 0.9966: the tie-break moves roughly 0.3% of candidates. The dependence is therefore real and we do not claim permutation invariance; its effect is far below the scale of any attribution difference the paper reports. A source-symmetric key — for instance the mean reciprocal rank across sources — would remove it, and we would adopt one in a revision of the design.

This choice is load-bearing. The obvious alternative — draw one candidate pool from the full fused scorer — is *invalid*, because it hands the grand coalition a retrieval

pool selected in its own favour. Every $v(S)$ for $S \subsetneq \mathcal{G}$ is then evaluated on items chosen to suit $\mathcal{G}$, so $v(\mathcal{G})$ is inflated relative to all of them. Since Shapley values are built entirely from differences $v(S \cup \{g\}) - v(S)$, that inflation contaminates every attribution. Under the union design each coalition scores an identical item set, so differences in $v(S)$ reflect *fusion within* S , not retrieval. We retain the rejected design as an ablation and quantify the bias it induces.

Candidate recall is a ceiling, not a detail

A test item outside C_u contributes zero to NDCG@10 by construction. Candidate recall $\Pr[\text{test}_u \in C_u]$ therefore upper-bounds every ranking metric downstream, and we report it before any attribution number (Table 1).

4.2 Characteristic function

Let $z_{u,g,i}$ be the per-user, per-source z -normalised score. For coalition S , $\hat{s}_{u,i}(S) = \sum_{g \in S} w_g^{(S)} z_{u,g,i}$ with $w^{(S)}$ a head fitted on the validation fold, and Let $y_{u,i} = \mathbb{1}[i = i_u^{\text{test}}]$ for $i \in C_u$ be the binary relevance vector, let $\pi_u(S)$ be the ranking of C_u induced by $\hat{s}_{u,\cdot}(S)$ under a deterministic tie rule (descending score, then ascending item index). Define the *per-user* baseline and game

$$b_u = \mathbb{E}_{\pi \sim \mathcal{U}(C_u)} [\text{NDCG@10}(\pi; y_u)], \quad v_u(S) = \text{NDCG@10}(\pi_u(S); y_u) - b_u, \quad v(S) = \frac{1}{|\mathcal{U}|} \sum_u v_u(S). \quad (3)$$

The baseline is the *expected* payoff of a uniformly random ranking, not a sampled one. With a single held-out item and IDCG = 1 it is available in closed form: the item is equally likely to land in any of the $|C_u|$ positions, so

$$b_u = \frac{1}{|C_u|} \sum_{r=1}^{\min(K, |C_u|)} \frac{1}{\log_2(r+1)}, \quad \text{and } b_u = 0 \text{ if } i_u^{\text{test}} \notin C_u. \quad (4)$$

If $i_u^{\text{test}} \notin C_u$ then $y_u = \mathbf{0}$ and $\text{NDCG@10}(\cdot; y_u) = 0$ for every ranking, so $v_u(S) = 0$; such users contribute zero to every coalition and are the reason candidate recall bounds all downstream metrics. Because $v_u(\emptyset)$ is defined as the expected random payoff and b_u is that same quantity, the empty coalition is zero per user and not merely on average, which is what Property 3 requires.

The coalition head is the ridge solution on the validation fold,

$$w^{(S)} = \arg \min_{w \in \mathbb{R}^{|S|}} \sum_u \frac{1}{|C_u|} \|\mathbf{y}_u^{\text{val}} - Z_u^{(S)} w\|_2^2 + \lambda \|w\|_2^2, \quad (5)$$

where $Z_u^{(S)} \in \mathbb{R}^{|C_u| \times |S|}$ stacks the z -normalised scores of the sources in S over $i \in C_u$, and $\mathbf{y}_u^{\text{val}} \in \{0, 1\}^{|C_u|}$ marks the user’s held-out validation interaction. The $1/|C_u|$ factor matters: candidate sets differ in size across users, and without it users with larger pools would dominate the fit. Weights are unconstrained and no intercept is fitted. Test interactions are never used to fit $w^{(S)}$.

Why the baseline is an expectation and not a draw

Earlier versions of this work used a single seeded permutation as π_u^0 . That was a mistake, and the reason is worth stating because it is easy to miss. The baseline cancels in $v_u(S \cup \{g\}) - v_u(S)$ whenever $S \neq \emptyset$, so it looks like a pure normalisation. It does not cancel at $S = \emptyset$: there the marginal is $v(\{g\}) - 0$, which retains b_u . Since the empty set carries Shapley weight $w(0) = 1/n$, changing the baseline shifts *every* φ_g by exactly $-\Delta/n$, where Δ is the change in the mean baseline.

We verified this rather than reasoning about it. Re-running the identical game with a different permutation seed moved the mean baseline by -0.000796 and shifted all five MovieLens values by $+0.00015926 = 0.000796/5$, agreeing to eight decimal places — and moved φ_{ct} across zero. A sign-based finding must not depend on which permutation happened to be drawn, so Equation (4) removes the draw entirely. Under the expectation, re-running with any seed reproduces every φ_g to 0 in exact arithmetic; we measure $0.00\text{e}+00$. Efficiency is unaffected either way, since $v(\mathcal{G})$ shifts with the baseline.

A residual, quantified source of cross-platform variation

Removing the sampled baseline removes the *gratuitous* randomness, not all numerical variation. Comparing a run on Apple silicon against one on x86-64, with identical code, configuration and seeds, the candidate sets agree in size exactly ($|C_u|$ mean 599.9985, s.d. 0.067, min 597, max 600 on both) but differ in *contents* for 2 of 6035 users — 0.03%. The cause is the BLAS backend: the ALS and truncated-SVD scorers accumulate in different orders on the two platforms, which flips a handful of near-ties at the truncation boundary. Downstream this moves $v(\mathcal{G})$ by 5×10^{-4} and the raw monotonicity count between 16 and 23, while the material count and every reported sign are unchanged.

We report this because it bounds what “exact” means in our title. The *aggregation* over the 2^5 coalitions is exact and carries no sampling error; the *game* it aggregates is a fitted object computed in floating point, and two conforming platforms need not produce bit-identical candidate sets. This is the reason we report a material monotonicity count alongside the raw one, and the reason the artefact records the platform and library versions of the run that produced it.

The regularisation strength is frozen

λ is chosen once on a pilot and never tuned per coalition. Per-coalition tuning is not benign: it is optimistic bias that *grows with* $|S|$, since larger coalitions have more parameters and more opportunity to overfit the held-out interaction. That would inflate $v(\mathcal{G})$ relative to small coalitions — the same pathology as a grand-coalition candidate pool, arriving by a different route.

4.3 Exact aggregation

$$\varphi_g = \sum_{S \subseteq \mathcal{G} \setminus \{g\}} \frac{|S|! (|\mathcal{G}| - |S| - 1)!}{|\mathcal{G}|!} [v(S \cup \{g\}) - v(S)], \quad (6)$$

an exact sum over all $2^5 = 32$ coalitions; for $n = 5$ the weights by $|S|$ are $(\frac{1}{5}, \frac{1}{20}, \frac{1}{30}, \frac{1}{20}, \frac{1}{5})$.

What “exact” means

The aggregation has *no sampling error*, unlike Monte-Carlo Shapley. But $v(S)$ is itself *fitted*, so φ_g carries estimation error. We write “exact given the fitted v ” throughout, and report seed-based confidence intervals in the main text rather than an appendix. Efficiency, by contrast, holds exactly regardless of fit noise because it is structural.

5 Algorithm and Cost

Algorithm 1 states the procedure end to end. Two details carry the correctness of everything downstream and are easy to get wrong. The candidate set is built *once*, from all sources, and reused unchanged for every coalition (line 4): building it per coalition would make $v(S)$ incomparable across the lattice, because a coalition would be scored on candidates it selected for itself. And the ridge head for each coalition is fit against *validation* targets while $v(S)$ is scored against *test* items (lines 9 and 12), so no coalition value is measured in-sample.

Cost

Let $n = |\mathcal{G}|$, m users, $N = |C_u|$, and d the item-embedding dimension. Source training is the usual cost of the five components and is independent of the game. Candidate construction is $O(mnN \log N)$. The step that matters is line 8: accumulating the Gram matrix once costs $O(mNn^2)$, after which each of the 2^n coalitions is an $O(n^3)$ solve on a slice plus $O(mN \log N)$ to rank and score. Total is

$$O(mNn^2 + 2^n(n^3 + mN \log N)). \quad (7)$$

The naive implementation rebuilds the design matrix per coalition, giving $O(2^n \cdot mNn^2)$; on MovieLens that difference was 81 minutes against 40 seconds. Memory is dominated not by the game but by the n dense score matrices, $4mn|\mathcal{I}|$ bytes in single precision — 129 GB at full Gowalla size, which is why the corpora are subsampled to a declared budget.

The 2^n factor is the binding constraint on scaling. At $n = 5$ exhaustive enumeration costs 32 refits and is unambiguously worth it; at $n = 20$ it would be 10^6 and sampling becomes necessary, at which point the paper’s central claim of an exact, variance-free aggregation no longer holds. We regard $n \lesssim 12$ as the practical range, which covers the architectural-source setting this paper addresses but not feature-level attribution.

Measured wall-clock for the full E0–E8 pipeline on an Apple M4 with 48 GB, single seed: MovieLens 163 s, Amazon-VG 71 s, Gowalla 943 s, of which the 32 coalition evaluations are 13, 10, and 285 s respectively. The game itself is never the bottleneck; candidate construction and the neural reference baselines dominate.

Algorithm 1 Exact ranking-stage SignalShap

Require: train $\mathcal{D}_{\text{tr}}$, validation $\mathcal{D}_{\text{val}}$, test $\mathcal{D}_{\text{te}}$; sources $\mathcal{G}$, $n = |\mathcal{G}|$; cap N_{max} ; penalty λ ; cutoff K ; baseline seed; tie tolerance ϵ

Ensure: coalition values $v(S)$, source values φ_g , per-user $\varphi_g(u)$

```
1: fit every source  $g \in \mathcal{G}$  on  $\mathcal{D}_{\text{tr}}$  only
2: score all eligible items per user; mask items seen in training
3: for each user  $u$  do
4:    $C_u \leftarrow$  truncated union of per-source top- $N$  lists ▷ once, all sources
5:    $Z_u \in \mathbb{R}^{|C_u| \times n} \leftarrow$   $z$ -normalised source scores over  $C_u$ 
6:    $\mathbf{y}_u^{\text{val}}, \mathbf{y}_u^{\text{te}} \in \{0, 1\}^{|C_u|} \leftarrow$  indicators of the validation and test items
7:    $\pi_u^0 \leftarrow$  permutation of  $C_u$  from hash(seed,  $u, \cdot$ )
8:    $b_u \leftarrow \text{NDCG@10}_K(\pi_u^0; \mathbf{y}_u^{\text{te}})$  ▷ baseline scored on TEST
9: end for
10:  $A \leftarrow \sum_u |C_u|^{-1} Z_u^\top Z_u$ ,  $c \leftarrow \sum_u |C_u|^{-1} Z_u^\top \mathbf{y}_u^{\text{val}}$  ▷ VALIDATION targets;
    accumulate once
11: for each  $S \subseteq \mathcal{G}$  do
12:   if  $S = \emptyset$  then
13:      $v_u(S) \leftarrow \text{NDCG@10}_K(\pi_u^0; \mathbf{y}_u^{\text{te}}) - b_u = 0$  for all  $u$ 
14:   else
15:     if  $\lambda > 0$  then
16:        $w^{(S)} \leftarrow [A_{SS} + \lambda I_{|S|}]^{-1} c_S$  ▷ slice  $A$  and  $c$ ; do not rebuild
17:     else
18:        $w^{(S)} \leftarrow A_{SS}^+ c_S$  ▷ minimum-norm;  $A_{SS}$  is singular under exact
        duplicates
19:     end if
20:      $v_u(S) \leftarrow \text{NDCG@10}_K(\text{rank}_\epsilon(Z_u^{(S)} w^{(S)}); \mathbf{y}_u^{\text{te}}) - b_u$  for all  $u$ 
21:   end if
22:    $v(S) \leftarrow \text{mean}_u v_u(S)$ 
23: end for
24: for each  $g \in \mathcal{G}$  do
25:    $\varphi_g \leftarrow \sum_{S \subseteq \mathcal{G} \setminus \{g\}} \frac{|S|! (n - |S| - 1)!}{n!} [v(S \cup \{g\}) - v(S)]$ 
26:    $\varphi_g(u) \leftarrow \sum_{S \subseteq \mathcal{G} \setminus \{g\}} \frac{|S|! (n - |S| - 1)!}{n!} [v_u(S \cup \{g\}) - v_u(S)]$ 
27: end for
28: verify efficiency  $\sum_g \varphi_g = v(\mathcal{G})$ , the per-user decomposition, and that no player is
    structurally degenerate
```

6 Formal Properties

Properties 1–3 are instantiations of standard Shapley axioms; we claim no novelty for their mathematical content, only for their careful use here.

Property 1 (Efficiency). $\sum_{g \in \mathcal{G}} \varphi_g = v(\mathcal{G})$.

Property 2 (LOO redundancy collapse). Let v be *monotone*, i.e. $v(S \cup \{g\}) \geq v(S)$ for all S and $g \notin S$. If g_1, g_2 satisfy $v(S \cup \{g_1\}) = v(S \cup \{g_2\}) = v(S \cup \{g_1, g_2\})$ for every

$S \subseteq \mathcal{G} \setminus \{g_1, g_2\}$, then $\text{LOO}(g_1) = \text{LOO}(g_2) = 0$ while $\varphi_{g_1} = \varphi_{g_2} \geq v(\{g_1\})/|\mathcal{G}| > 0$ whenever $v(\{g_1\}) > 0$.

The monotonicity hypothesis is not optional

It is tempting to state Property 2 without it. That statement is **false**. Consider three players with $v(\emptyset) = 0$, $v(g_1) = v(g_2) = 1$, $v(g_3) = 5$, $v(g_1g_2) = 1$, $v(g_1g_3) = v(g_2g_3) = 0.5$, and $v(\mathcal{G}) = 0.5$. Redundancy holds exactly, $v(\{g_1\}) = 1 > 0$, and $\text{LOO}(g_1) = \text{LOO}(g_2) = 0$ as predicted. Symmetry and efficiency both hold. Yet

$$\varphi_{g_1} = \varphi_{g_2} = -0.4167, \quad \varphi_{g_3} = +1.3333, \quad \sum = 0.5 = v(\mathcal{G}).$$

The redundant sources receive *negative* credit despite positive standalone value. Restoring monotonicity repairs it: on the monotone variant ($v(g_1g_3) = v(g_2g_3) = v(\mathcal{G}) = 5.5$) one obtains $\varphi_{g_1} = 0.4167 \geq v(\{g_1\})/3 = 0.3333$. Both games are encoded as regression tests in the artefact.

This matters practically: the coalition-conditional refit is precisely the mechanism that can break monotonicity, since a head fitted on $S \cup \{g\}$ may score worse than one fitted on S . We therefore *audit* monotonicity on all $|\mathcal{G}| \cdot 2^{|\mathcal{G}|-1} = 80$ pairs rather than assume it, and report violations as findings.

Property 3 (Free per-user decomposition). $\varphi_g = \frac{1}{|\mathcal{U}|} \sum_u \varphi_g(u)$, by linearity of the Shapley operator.

Lemma 1 (ε -substitutability). *Exact redundancy is a measure-zero condition that is unlikely to hold on data without controlled duplication, so Property 2 motivates but cannot instantiate an empirical test. We therefore use a two-sided condition. Call g_1, g_2 ε -**substitutable** if, for every $S \subseteq \mathcal{G} \setminus \{g_1, g_2\}$,*

$$\max\{|v(S \cup \{g_1\}) - v(S \cup \{g_1, g_2\})|, |v(S \cup \{g_2\}) - v(S \cup \{g_1, g_2\})|\} \leq \varepsilon. \quad (8)$$

Then:

- (i) $|\text{LOO}(g_1)| \leq \varepsilon$ and $|\text{LOO}(g_2)| \leq \varepsilon$.
- (ii) If in addition v is monotone and $v(\{g_1\}) \geq c$, then $\varphi_{g_1} \geq c/|\mathcal{G}|$, with no ε correction.

Proof. (i) Taking $S = \mathcal{G} \setminus \{g_1, g_2\}$ in (8) gives $|v(\mathcal{G} \setminus \{g_2\}) - v(\mathcal{G})| \leq \varepsilon$ and $|v(\mathcal{G} \setminus \{g_1\}) - v(\mathcal{G})| \leq \varepsilon$, which are exactly $|\text{LOO}(g_2)| \leq \varepsilon$ and $|\text{LOO}(g_1)| \leq \varepsilon$. The two-sided form of (8) is what makes both statements available; a one-sided condition bounds only one of them.

(ii) Write the Shapley value in permutation form, $\varphi_{g_1} = \sum_{T \subseteq \mathcal{G} \setminus \{g_1\}} w(|T|) [v(T \cup \{g_1\}) - v(T)]$ with $w(t) = t!(n-t-1)!/n!$ and $n = |\mathcal{G}|$. Under monotonicity every summand is non-negative, so discarding all terms except $T = \emptyset$ can only decrease the total:

$$\varphi_{g_1} \geq w(0) [v(\{g_1\}) - v(\emptyset)] = \frac{1}{n} v(\{g_1\}) \geq c/n.$$

Two remarks. First, the bound in (ii) follows from monotonicity alone and does *not* require substitutability; we state the two parts together only because the pair is what makes the LOO-Shapley gap quantitative. Second, the anchor weight is $w(0) = 1/n$,

not the mass $\sum_{T \subseteq \mathcal{G} \setminus \{g_1, g_2\}} w(|T|) = 1/2$ that corresponds to $\Pr[g_1 \text{ precedes } g_2]$ under a uniform random ordering. Conflating the two would overstate the bound by a factor of $n/2$. $\square$

Combining the two parts, an ε -substitutable pair in a monotone game has a LOO-Shapley gap of at least $c/n - \varepsilon$ for each member. This is the quantitative form of the qualitative statement in Property 2.

Two corrections worth recording. The constant in Lemma 1(ii) is $1/|\mathcal{G}|$, *never* $1/2$: substituting the total weight mass for the anchor weight overstates the bound by $n/2$, a factor of 2.5 at $n = 5$, and is unattainable — on the monotone game above $\frac{1}{2}v(\{g_1\}) = 0.5 > 0.4167 = \varphi_{g_1}$. Relatedly, no pairwise closed form $\frac{1}{2}[v(\mathcal{G}) - v(\mathcal{G} \setminus \{g_1, g_2\})]$ is valid; it returns -2.25 against a true -0.4167 on the counterexample.

Remark 1 (Positive-affine scale invariance). *Per-user z -normalisation makes φ_g invariant under user- and source-specific maps $z \mapsto az + b$ with $a > 0$. The restriction matters: a negative a reverses the induced ranking and is not an invariance, and the normalisation is undefined for a constant score vector, which we handle by mapping zero-variance sources to the zero vector. Invariance does not extend to nonlinear monotone maps.*

7 Experimental Setup

Players

cf: implicit-feedback ALS [18], 64 factors. *ct*: TF-IDF cosine over item metadata. *pop*: time-decayed global frequency. *rec*: exponential recency over an item’s content cluster. *seq*: item2vec, realised as SVD of the shifted PPMI co-occurrence matrix. Levy and Goldberg [23] show that skip-gram with negative sampling implicitly factorises a shifted PMI matrix; we factorise it directly, which is deterministic on CPU. We call this a shifted-PPMI SVD sequential model rather than claiming exact algorithmic equivalence to skip-gram.

Expected overlap, stated in advance

Two pairs overlap *by construction* and we state this before reporting results rather than present it as a discovery: *pop-cf* (ALS on implicit feedback chases popularity) and *rec-ct* (recency is defined over content clusters). RQ2 asks whether Shapley *recovers* known structure — a validation of the estimator against a predeclared qualitative expectation.

Corpora

Three real datasets spanning a $36\times$ density range (Table 1): MovieLens-1M [12], Gowalla check-ins [4], and the Video Games category of the Amazon Reviews corpus [17, 27]. All three carry genuine interaction timestamps, which is a requirement rather than a convenience: three of the five players (*rec*, *seq*, and the decay term of *pop*) are uninterpretable without them.

We record a withdrawal. Earlier drafts of this work used the Yelp2018, Gowalla, and Amazon-Book splits distributed with LightGCN [14], chosen so that preprocessing

would match published practice. Those splits discard timestamps, and our first runs substituted file order — which is not a weak proxy for time but an arbitrary one. We replaced them with the original timestamped sources and report only the latter. The affected artefacts are retained in the repository, marked as a pilot, so the substitution is auditable rather than merely asserted.

Gowalla and Amazon are subsampled to fit a memory budget, with per-corpus scale *derived* from that budget rather than chosen: five dense score matrices at the full Gowalla size would require roughly 129 GB.

Content for ct and rec

Neither Gowalla nor Amazon ships item text in the form the ct scorer expects, and our first runs on them left both players with no input at all. The consequence was not a visible crash but a silent one: ct returned an all-zero matrix, rec collapsed to a single content cluster, and both received Shapley values indistinguishable from zero. Read naively, two players looked empirically worthless when they had simply never been supplied. We therefore derive content from what each corpus does carry — Gowalla venues from their coordinates, as nested geographic cells at roughly 100, 10, and 1 km (each emitted on two half-offset grids, so that venues either side of a cell boundary still share a token); Amazon items from the category path and store fields of the product-metadata dump. A structural audit now runs before any coalition is scored and refuses to proceed when a player cannot reorder any candidate list (Section 8).

Protocol

Leave-last-out temporal split where timestamps exist; ties broken by (timestamp, record index). The main study uses seeds {42, 43, 44}; the seed-interval and equivalence analyses of Appendix A use ten seeds (42–51), which is what the two-level bootstrap resamples over. Wilcoxon signed-rank [39] for paired NDCG@10, permutation tests (10 000 shuffles) for gaps and heterogeneity, Holm–Bonferroni [16] with family size *and composition* declared, Cohen’s d_z throughout.

Reporting protocol

All ranking comparisons evaluate *every* method over the full catalogue; our method scores items outside C_u as $-\infty$ so its recall ceiling is a visible cost rather than a hidden denominator advantage. Restricting the baseline to C_u would be disclosable but is simply a weaker experiment. Note the two-protocol split: $v(S)$ remains fixed-candidate by definition of the game; only reporting is full-catalogue.

8 Preconditions

Table 1 carries the two gates, and on this corpus set both bite.

Monotonicity

On all three corpora the fitted game is **non-monotone**, so Property 2 does not apply as stated and a negative φ_g must be read as substantive rather than as numerical error. We regard this as a genuine finding about coalition-conditional refitting rather than

Table 1 Preconditions per corpus. Candidate recall bounds every ranking metric that follows; the monotonicity column decides whether Property 2 applies at all. Both must be inspected before any attribution number is interpreted. Density is computed as-used, never quoted from a source paper. Corpora marked *ceiling* are reported under the restriction stated in the text: relative contrasts only.

Corpus	Users	Items	Density	$N_{\max}$	Recall	Monot. (mat./raw)	Prop. 2
MovieLens-1M	6 038	3 533	2.697%	600	0.748	7 / 23	no
Amazon-VG	7 120	3 516	0.469%	600	0.588 [†]	4 / 12	no
Gowalla	8 865	82 134	0.075%	11 623	0.450 [†]	6 / 13	no

[†]Below the 0.60 gate; reported under the ceiling restriction described in this section (relative contrasts only).

an inconvenience: adding a source to a coalition can make the refitted ridge weights worse for some users.

Table 1 reports two counts, and the distinction is not pedantry. The raw count is every audited pair with $v(S \cup \{g\}) < v(S)$; the material count keeps only those with $|v(S \cup \{g\}) - v(S)| > 10^{-3}$. Most violations are far below the scale of v itself — the median is 4.5×10^{-4} against $v(\mathcal{G}) \approx 0.05$ — so the raw count is sensitive to floating-point ordering and is not a reproducible statistic. We learned this concretely: two machines running identical code on identical data reported 17 and 23 raw violations while both reported the same material count. The Property 2 argument is what we quote as the headline.

To be precise about what each count licenses, since these are three different things. *Exact* monotonicity is the theorem’s hypothesis, and it fails on all three corpora at any tolerance — Property 2 does not apply, full stop, and no threshold recovers it. The *material* count answers a different question: whether the violations are large enough to matter for interpretation, which bears on how seriously to read a negative φ_g . The *raw* count answers neither reliably, because it is not reproducible. We report material and raw, use exact for the theorem, and do not use the threshold to weaken a mathematical hypothesis. It is the reason the monotonicity hypothesis was made explicit in Section 6 rather than assumed.

Candidate recall, and a limit we can locate

MovieLens clears the 0.60 gate at 0.748. Neither other corpus does, and the two fail for different reasons that we did not treat alike.

For Amazon-VG, recall is 0.588 at $N_{\max} = 600$. That value is 17.1% of its catalogue, the same fraction MovieLens receives; a log fit to the measured recall curve puts the 0.60 crossing at $N_{\max} \approx 634$. Raising $600 \rightarrow 800$ would clear the gate. We did not, and the reason is methodological rather than technical: $N_{\max}$ is frozen in advance, and moving it by six percent of a threshold *after* observing that threshold is selection on the very quantity the parameter gates. The corpus is retained under the ceiling restriction instead.

For Gowalla the situation is qualitatively different, and more interesting. We did raise $N_{\max}$, from 2 500 to 11 623 — the first value was inherited from a differently sized

split and amounted to only 3.04% of this corpus’s catalogue, against 17% elsewhere, so the raise is justified by consistency rather than by outcome. We note for exactness that 11 623 was derived from a 68 443-item extract of this corpus and is 14.2% of the 82 134-item version finally used — close to, but not identical with, the 17% the rule prescribes. We did not re-derive it after the corpus grew, precisely because adjusting $N_{\max}$ once recall is known is the selection we are trying to avoid. Recall improved to 0.450. The robustness sweep then measured the curve directly: $N_{\max} = 5\,811 \rightarrow 0.418$, $11\,623 \rightarrow 0.450$, $23\,246 \rightarrow 0.471$. Doubling the pool buys +0.021.

We are deliberately careful about what this does and does not establish. Recall is *not* bounded below one: every unseen item receives a finite score from every source, so setting $N_{\max}$ to the catalogue size makes C_u the whole eligible catalogue and recall approaches one by construction. We verified this on MovieLens, where $N_{\max} = 3\,533$ yields recall 0.9995. An earlier version of this manuscript extrapolated the logarithmic fit beyond the sampled range and concluded that Gowalla’s gate was unreachable at any pool size. That conclusion was wrong — the logarithm describes the curve locally and cannot hold asymptotically — and we withdraw it.

What the measurements do establish is a cost. On Gowalla, clearing 0.60 would require a candidate set large enough that the fixed-candidate design stops being a retrieval-plus-ranking decomposition and becomes full-catalogue ranking, at which point the game no longer models the two-stage architecture it was built to explain, and the 32 coalition refits become correspondingly expensive. The practical statement is therefore about affordability and design validity rather than impossibility: on a corpus this sparse, the candidate pool needed for a usable recall ceiling is a large fraction of the catalogue, and a practitioner should check that ratio before adopting a fixed-candidate game. Quantifying where that tradeoff becomes unacceptable, and whether recall rises fast enough in the unsampled region between 23 246 and 82 134 to change the assessment, requires a full-catalogue sweep we have not run.

What the ceiling restriction means

Both affected corpora are retained for *relative* comparisons — the LOO-versus-Shapley contrast of Section 11, which is a difference between two attributions computed on the same candidate sets — and excluded from absolute NDCG@10 comparison against the neural baselines, where a suppressed recall ceiling would flatter or penalise methods unequally. This follows the fallback ladder fixed before these runs. The restriction is enforced in code: an artefact whose gate failed is admitted only if its corpus appears in a declared exemption list, and it carries the restriction string into every table built from it.

Per-source attributions with seed-based intervals appear in Figure 2. Efficiency holds to machine precision on every corpus, with worst-case $|\sum_g \varphi_g - v(\mathcal{G})| = 6.9 \times 10^{-18}$.

8.1 Configuration

Table 2 lists every value that affects a reported number. All are fixed in a version-controlled configuration file before the runs and are not tuned per corpus or per coalition; where a value was amended after the first runs, the file records the date and

Table 2 Complete configuration. Values marked $\dagger$ were selected once on a MovieLens pilot and then frozen; the remainder are defaults never varied. $N_{\max}$ is the only per-corpus value

Component	Parameter	Value	Note
<i>cf</i>	latent factors	64	ALS, implicit feedback
	regularisation	0.1	
	alternating iterations	15	
<i>ct</i>	TF-IDF vocabulary	5000	token pattern \S+
<i>pop</i>	decay half-life	180 d	exponential
<i>rec</i>	content clusters	20	truncated SVD of TF-IDF
	decay half-life	30 d	
<i>seq</i>	embedding dimension	64	shifted-PPMI SVD
	co-occurrence window	5	
Game	$N_{\max}$	600 / 600 / 11 623	ML-1M / Amazon-VG / Gowalla frozen, never per coalition
	ridge λ	$1.0^\dagger$	
	NDCG cutoff K	10	
	baseline seed	42	fixes π_u^0
	candidate growth iters	10	
	recall gate	0.60	see Section 8
	materiality threshold	10^{-3}	reporting only, not a theorem
Fusion	segments	4	activity $\times$ recency, median splits
	softmax τ	$\{0.001, 0.01, 0.1\}^\dagger$	validation-selected
	shrinkage ρ	$\{0, 0.5, 1\}^\dagger$	validation-selected
LightGCN	factors / layers	64 / 3	contextual reference only
	epochs / lr / reg	40 / 0.01 / 10^{-4}	
SASRec	dim / max length	64 / 50	contextual reference only
	blocks / epochs / lr	2 / 25 / 0.05	
Protocol	seeds	42–44 (main), 42–51 (intervals)	test = last, validation = second-last
	split	leave-last-out	
	tie-break	(timestamp, record index)	

the reason. We report them in the paper rather than only in the repository, because a frozen configuration is not a substitute for a stated one.

Preprocessing and subsampling

MovieLens-1M is binarised at rating ≥ 4 , which is why the as-used interaction count is 575 276 rather than one million. Gowalla and Amazon-VG are filtered to a 10-core and binarised (Amazon at rating ≥ 4 ; Gowalla check-ins are implicit). Both are then subsampled by drawing users uniformly at random without replacement under seed 42, retaining all interactions of the drawn users; the number of users is *derived* from a declared memory budget rather than chosen, since five dense score matrices at $8\,865 \times 82\,134$ already occupy 14.6 GB and the full corpus would need 129 GB. Item vocabularies are re-indexed after subsampling, so the catalogue sizes in Table 1 are post-subsampling.

9 Validation Against Known Axiomatic Constraints

An attribution method should be judged by whether it returns the right answer, not only by whether its output looks plausible. Yet on observational data the right answer is unknown, which is why attribution methods are so rarely validated directly. This section supplies a known *axiomatic constraint* by construction, on real corpora, and shows that SignalShap satisfies it. We avoid the term ground truth: duplicate injection fixes what the attribution must satisfy, not what the correct magnitudes are.

Design

We inject a near-duplicate of an existing source, $g_{\text{dup}} = g + \eta \sigma_g \epsilon$ with ϵ standard normal, and re-run the game with six players. The candidate set is held *fixed* at its pre-injection value, so the only thing that changes is the player set.

What is known in advance, and what is not

At $\eta = 0$ the original and duplicated sources are exchangeable in the declared game, so any rule satisfying the symmetry axiom must assign them equal values. This gives a checkable constraint that does not depend on the data. Property 2 suggests a second outcome: leave-one-out should collapse toward zero for *both* members, since either alone is dispensable. **This one is not structurally guaranteed under our fitted game, and we do not treat it as such.** Duplicating a column in a ridge regression is not prediction-invariant: two identical columns share the penalty, so the effective regularisation on their common direction is halved and the fitted scores can move. Exchangeability of the pair is guaranteed; redundancy of the coalition *values* is an empirical question. We therefore measure it, reporting

$$\Delta_{\text{red}} = \max_{S \subseteq \mathcal{G} \setminus \{g, g_{\text{dup}}\}} \max\{|v(S \cup \{g\}) - v(S \cup \{g, g_{\text{dup}}\})|, |v(S \cup \{g_{\text{dup}}\}) - v(S \cup \{g, g_{\text{dup}}\})|\} \quad (9)$$

alongside the symmetry error, and we run the diagnostic at both the frozen λ and at $\lambda = 0$. The latter matters because ordinary least squares *is* invariant to a repeated column — with the minimum-norm solution a shared coefficient splits evenly and predictions are unchanged — so $\lambda = 0$ turns the redundancy prediction into a structural guarantee and isolates whether any deviation at the frozen λ is due to regularisation.

We are deliberately precise about the limits of this design. **Duplication does not conserve credit**, and it would be wrong to require that it should. The combined Shapley value of a cloned pair need not equal the original player’s pre-duplication value: in the two-player monotone game $v(g) = 7$, $v(h) = 8$, $v(g, h) = 8$ one has $\varphi_g = 3.5$, whereas replacing g by perfect substitutes g_1, g_2 gives $\varphi_{g_1} = \varphi_{g_2} = 7/3$ and a pair sum of $14/3 \approx 4.67$. Clone insertion redistributes credit among *all* players, a sensitivity of Shapley allocations to replication that is known in the cooperative-game literature [11]. Our own runs show the same effect: on MovieLens-1M the pair sum is 0.02208 against a pre-injection $\varphi_{cf} = 0.01547$, a ratio of 1.43. We therefore treat the pair sum as a *diagnostic*, not a requirement, and the experiment as a symmetry and implementation check rather than as recovery of absolute credit.

Table 3 Duplicate-source diagnostics on MovieLens-1M (injected clone of cf , $\eta = 0$) at the frozen ridge penalty and at $\lambda = 0$. Δ_{red} is the coalition-level redundancy deviation of Eq. (9), measured over all $2^4 = 16$ background coalitions rather than assumed. Symmetry is guaranteed by exchangeability; redundancy is not, and is reported as an empirical outcome

Corpus	λ	Sym. err.	Δ_{red}	LOO(cf)	LOO(cf_{dup})
MovieLens-1M	frozen	0.0	0.0	0.000000	0.000000
	0	0.0	0.0	0.000000	0.000000

Table 3 reports the outcome. The result is unambiguous and reproducible: Shapley satisfies symmetry exactly in every condition. LOO returns **exactly zero** for both members of every pair — not approximately, exactly — because $v(\mathcal{G}) = v(\mathcal{G} \setminus \{g\})$ holds numerically when a perfect substitute remains in the coalition.

The practical reading is stark. An engineer running standard ablation on a system containing two redundant components is told that *neither contributes anything*. Acting on that, they would retire both and lose the capability entirely. This is the failure mode the paper is about, and here it is measured rather than argued.

Graceful degradation

As η grows the duplicate becomes a noisier copy and exact redundancy weakens. LOO becomes non-zero but erratic in sign, which is the behaviour Lemma 1(i) describes for ε -redundancy.

10 Validating the Aggregation

10.1 Analytic games with known Shapley vectors

Duplicate injection tests symmetry, but symmetry alone does not validate *magnitudes*: a method that halved every value would still pass it. We therefore also evaluate six analytic games whose Shapley vector is known in closed form, chosen to span the structures our real games might exhibit (Table 4). These are unit tests of the aggregation, not evidence about recommender data; they establish that the enumeration in Equation (4) is implemented correctly, which is a precondition for interpreting anything computed with it.

We additionally verify *additivity*, the remaining uniqueness axiom that no other diagnostic here exercises: efficiency is a sum constraint, symmetry is covered by duplicate injection, and the null player has its own game. Summing an additive and a unanimity game — structurally different, so the check is not vacuous — gives $\varphi(v_1 + v_2) = \varphi(v_1) + \varphi(v_2)$ to 4.4×10^{-16} .

The null-player and harmful games matter most for this paper. Our fitted games are non-monotone and produce negative φ_g (Section 11), so it is necessary to establish that a negative value is a faithful computation rather than an artefact of the implementation; the harmful game confirms that -2 is recovered exactly. The null-player

Table 4 Analytic games with known Shapley vectors, each defined by an explicit characteristic function so the check is reproducible from the paper alone. $\mathbb{I}[\cdot]$ is the indicator. All six reproduce with zero error, which validates magnitudes, the null-player axiom, and the handling of negative contributions — none of which the symmetry diagnostic can detect. Additivity is verified separately by summing the additive and unanimity games.

Game	Characteristic function $v(S)$	Expected φ	Max error
Additive	$\sum_{g \in S} c_g, \quad c = (3, 2, 1)$	$(3, 2, 1)$	0
Substitutes	$4 \cdot \mathbb{I}[\{s_1, s_2\} \cap S \neq \emptyset] + \mathbb{I}[x \in S]$	$(2, 2, 1)$	0
Complements	$6 \cdot \mathbb{I}[\{c_1, c_2\} \subseteq S] + 2 \cdot \mathbb{I}[x \in S]$	$(3, 3, 2)$	0
Null player	$\sum_{g \in S \cap \{a, b\}} c_g, \quad c = (3, 2)$	$(3, 2, 0)$	0
Harmful	$5 \cdot \mathbb{I}[a \in S] - 2 \cdot \mathbb{I}[bad \in S]$	$(5, -2)$	0
Unanimity	$9 \cdot \mathbb{I}[S = \mathcal{G}]$	$(3, 3, 3)$	0

game confirms that a source contributing nothing receives exactly zero, which is the property the structural audit of Section 7 relies on.

10.2 Alternative values and pairwise interactions

Other semivalues

The Shapley value is one member of a family. We also compute the Banzhaf value, which weights every coalition equally rather than by permutation count, and a uniform semivalue, whose weights are $\psi_i = \frac{1}{n} \sum_{k=0}^{n-1} \binom{n-1}{k}^{-1} \sum_{|S|=k, S \subseteq \mathcal{G} \setminus \{i\}} [v(S \cup \{i\}) - v(S)]$, i.e. equal weight on each coalition *size* rather than on each coalition. On MovieLens all three induce the *same* source ordering (Kendall $\tau = 1.00$ between Shapley and Banzhaf, same top source). We computed this on MovieLens only, so the claim is that the MovieLens conclusions are robust to the choice among these values; we do not extend it to the other two corpora, where it remains untested. Banzhaf does not satisfy efficiency, so its values do not sum to $v(\mathcal{G})$ and only the induced ordering is comparable.

Pairwise interactions

Main-effect values cannot by themselves distinguish a source of low value from a source whose value is absorbed by a redundant partner. We therefore compute a pairwise interaction index: the classical *Shapley interaction index* of Grabisch and Roubens [10]. We flag a correction here. An earlier version of this manuscript reported these values as the order-2 Shapley–Taylor index [36] while computing the Grabisch–Roubens coefficient. The two are not interchangeable — at $n = 5$ their coalition weights differ by up to a factor of 2.5 — so the label was wrong even though the computation was a valid interaction index. We correct the label rather than the numbers, and note that Grabisch–Roubens does not satisfy Shapley–Taylor’s interaction-efficiency axiom: the pairwise terms do not sum with the singletons to $v(\mathcal{G})$. That is acceptable here because we use them only to *rank* redundancy, which is all the argument requires. Each pair

Table 5 Pairwise Shapley interaction indices (Grabisch–Roubens) on MovieLens-1M, seed 42. Negative entries indicate redundancy. One of the two most negative pairs, *cf-pop*, is a declared expected overlap; the other, *cf-seq*, is not, and the second declared overlap *rec-ct* is slightly positive

Pair	$\mathcal{I}_{ab}$	Pair	$\mathcal{I}_{ab}$
<i>cf seq</i>	−0.0561	<i>ct pop</i>	−0.0033
<i>cf pop</i>	−0.0256	<i>ct seq</i>	+0.0008
<i>pop rec</i>	−0.0015	<i>ct rec</i>	+0.0002
<i>cf rec</i>	−0.0013	<i>cf ct</i>	+0.0023
<i>rec seq</i>	−0.0011	<i>pop seq</i>	+0.0058

$\{a, b\}$ receives

$$\mathcal{I}_{ab} = \sum_{S \subseteq \mathcal{G} \setminus \{a, b\}} \frac{|S|!(n - |S| - 2)!}{(n - 1)!} [v(S \cup \{a, b\}) - v(S \cup \{a\}) - v(S \cup \{b\}) + v(S)], \quad (10)$$

the averaged second-order discrete derivative. A negative $\mathcal{I}_{ab}$ means the pair jointly contributes less than the sum of their separate contributions — the signature of redundancy — and a positive value indicates complementarity.

Table 5 localises the redundancy behind the MovieLens sign flip. The two most negative pairs are *cf|seq* (−0.056) and *cf|pop* (−0.026): collaborative filtering overlaps most with the sequential and popularity sources, which is why removing it from the grand coalition costs so little and why leave-one-out scores it below zero. Only one of these, *cf|pop*, was declared in advance (Section 7); *cf|seq* was not, and the other declared overlap *rec|ct* comes out slightly *positive* at +0.0002. We therefore claim partial rather than complete recovery of the pre-declared structure, and note that an earlier version of this paper overstated it. The interaction values are reported for MovieLens only; extending them to the other corpora, and attaching uncertainty, is straightforward but not yet done.

11 Attribution on Real Systems

Table 6 gives the per-source comparison and Figure 3 plots it. The result replicates in kind on all three corpora and we are careful to say exactly what replicates.

What is consistent, and what is not

Two of the three corpora contain a *material* sign disagreement — one where the gap exceeds 10^{-3} , the same threshold we apply to monotonicity. On MovieLens it is *cf*: Shapley scores it +0.01526, the second most valuable source in the system, where ablation scores it −0.00501 and marks it as worth removing. The gap is 0.0203, or

Table 6 LOO versus Shapley on all three corpora (seed 42). Sign disagreements are marked *material* when the gap exceeds 10^{-3} , the same threshold used for monotonicity; negligible flips are those where both estimates sit within noise of zero. Two of three corpora contain a material disagreement. The LOO, Shapley and Gap columns are seed 42, so that the gap is a like-for-like difference on one fitted game; the interval is over ten seeds and is reported here rather than only in the artefact. Under the deterministic baseline no interval spans zero on any corpus: every source’s sign is stable across seeds, including the two near-zero cases (φ_{ct} on MovieLens at -0.00018 and φ_{rec} on Amazon-VG at -0.00049) whose signs were not stable under the sampled baseline.

Corpus	Source	LOO	Shapley	95% CI (10 seeds)	Gap	
MovieLens-1M	<i>cf</i>	−0.00519	+0.01545	[+0.01547, +0.01602]	+0.02064	flip, material
	<i>ct</i>	+0.00044	−0.00009	[−0.00024, −0.00012]	−0.00053	
	<i>pop</i>	+0.00065	+0.00637	[+0.00639, +0.00652]	+0.00572	flip, negligible
	<i>rec</i>	−0.00005	+0.00026	[+0.00015, +0.00034]	+0.00031	
	<i>seq</i>	+0.01635	+0.02937	[+0.02966, +0.03018]	+0.01302	
Amazon-VG	<i>cf</i>	+0.00845	+0.01926	[+0.01943, +0.02016]	+0.01081	flip, negligible
	<i>ct</i>	+0.00002	+0.00187	[+0.00184, +0.00205]	+0.00185	
	<i>pop</i>	+0.00308	+0.00157	[+0.00167, +0.00189]	−0.00152	
	<i>rec</i>	+0.00023	−0.00047	[−0.00052, −0.00046]	−0.00070	
	<i>seq</i>	+0.00758	+0.01909	[+0.01868, +0.01898]	+0.01151	
Gowalla	<i>cf</i>	+0.00729	+0.00711	[+0.00632, +0.00652]	−0.00018	flip, material
	<i>ct</i>	+0.00528	+0.00674	[+0.00956, +0.00966]	+0.00146	
	<i>pop</i>	−0.00151	+0.00055	[+0.00086, +0.00099]	+0.00206	
	<i>rec</i>	+0.00001	+0.00038	[+0.00031, +0.00039]	+0.00037	
	<i>seq</i>	+0.00070	+0.00213	[+0.00355, +0.00366]	+0.00143	

39% of $v(\mathcal{G})$. On Gowalla it is *pop*, at 12.2% of $v(\mathcal{G})$: ablation scores the popularity signal negative where Shapley scores it positive.

Amazon-VG has no material disagreement. Its only sign flip, on *rec*, is 0.0008 — both estimates within noise of zero. We state this plainly because an earlier version of this manuscript claimed that *every* system contains a sign disagreement, and after correcting a reproducibility defect in the empty-coalition baseline (Section 15) that claim no longer holds. The honest statement is weaker: material sign disagreement between the two rules occurs on some but not all systems, it is not rare, and nothing in either rule announces in advance whether a given system will exhibit it.

The disagreement is not a regularisation artefact

A natural objection is that a sign flip could be an artefact of the frozen ridge penalty. We swept λ over eight orders of magnitude, 10^{-5} to 10^3 , recomputing the full game at each value. The material flips are invariant: *cf* flips on MovieLens and *pop* flips on Gowalla at *every* λ tested, and they are the only material flips on those corpora throughout. Amazon-VG has none until $\lambda = 10^3$, where the head is so heavily penalised that *ct* acquires one; we regard that endpoint as degenerate rather than informative. The disagreement therefore reflects coalition structure, not the regularisation regime.

One robustness check is available for the Gowalla result. That corpus was run at two subsample sizes, 5910 and 8865 users, drawn to different memory budgets. The

material *pop* disagreement is present at both, at 9.9% and 12.2% of $v(\mathcal{G})$ respectively, and *pop* is the only material flip in each case. A $1.5\times$ change in corpus size therefore does not create or destroy it. We report the larger run throughout.

The corollary is that a single-corpus reading would have been actively misleading. On MovieLens alone one would conclude that collaborative filtering is the source ablation systematically undervalues. Gowalla’s material flip falls on *pop* instead, and Amazon-VG has none. The results are consistent with the *mechanism* that disagreement tracks redundancy, though the interaction analysis supports that reading directly only on MovieLens. What certainly does not generalise is the identity of the affected source, or the guarantee that one exists at all.

Mechanism

The account is *qualitatively* the one Property 2 describes, though the property does not formally apply: all three fitted games are non-monotone (Table 1), so the positivity conclusion is unavailable and what follows is interpretation rather than theorem. A source whose signal is covered by others costs little when removed from the *grand* coalition, because the others absorb its role, and ablation observes only that single contrast. SignalShap averages the source’s contribution over all 16 coalitions in which it can appear, including those where nothing covers for it, and so recovers value that the overlap conceals. On MovieLens the collaborative signal overlaps with the sequential and popularity signals; on Gowalla it is the popularity signal that is covered, by a geographic content signal strong enough to rank second overall.

Ranking stability

The per-corpus orderings are $seq \succ cf \succ pop \succ rec \succ ct$ (MovieLens), $cf \succ seq \succ ct \succ pop \succ rec$ (Amazon-VG), and $cf \succ ct \succ seq \succ pop \succ rec$ (Gowalla). Pairwise Kendall τ between corpora is 0.40 (MovieLens–Amazon), 0.20 (MovieLens–Gowalla) and 0.80 (Amazon–Gowalla), none significant at $n = 5$ sources. We therefore make no cross-corpus consistency claim. An earlier version of this analysis reported τ between 0.60 and 0.80; that apparent agreement was an artefact of *ct* and *rec* having received no input on two corpora, which gave them identical near-zero values that agreed trivially. The corrected figures are lower and we report them in place of the finding they displaced.

Measured redundancy is associated with the attribution shift at $r = 0.525$ on MovieLens across the ten source pairs, in the direction the redundancy argument suggests. We are careful not to overstate this. Lemma 1 defines redundancy through *coalition values*, whereas this statistic is a Kendall correlation between per-user *score vectors*; the two are related but not equivalent, and the lemma predicts no particular correlation coefficient. The observation is consistent with the theory rather than a test of it, and with ten pairs it carries little inferential weight on its own.

Figure 4 shows the pairwise structure directly.

12 Limitations

Analytic and data-independent

The counterexample refuting unconditional Property 2; the $1/|\mathcal{G}|$ constant in Lemma 1; the bias argument against grand-coalition candidate pools; the bias argument against per-coalition λ tuning. These are mathematical and depend on no corpus.

Scope was narrowed after the study began

This work was designed as a study contrasting attribution across densities. The *hypotheses* were not changed to fit results: Properties 1–3 and Lemma 1 are corrected during revision, and both outcomes that went against expectation — the non-monotone fitted game and the inconclusive fusion comparison — are reported rather than removed. The intervention of Section 9 is **post-hoc in origin**: it was added because the observational comparison could not establish which attributor was correct. Its prediction was fixed by construction *before* it was run.

Temporal semantics: a flaw we removed rather than disclosed

An earlier version of this protocol carried a serious defect, and we describe it because the fix changed the corpus set. The LightGCN splits we first used carry no timestamps, so the recency player, the time-decay term of the popularity player, and the sequential player were operating on file order — an ordering with no temporal meaning at all. Attributions for *rec* and *seq* on those corpora were not small, nor lower bounds as a still earlier draft claimed; they were uninterpretable, because an arbitrary order can inflate or deflate a measured contribution in either direction.

The remedy was not a caveat but a replacement. We rebuilt the study on corpora that carry genuine interaction times, verifying for each that timestamps span a plausible range and are not degenerate: Gowalla spans 596 days with 99.2% distinct values, Amazon-VG spans 8 360 days with 97.4%. All five players now have the semantics the specification assigns them on all three corpora.

Two residual limitations remain. First, the content signal for *ct* and *rec* is derived rather than native on two corpora — geographic cells for Gowalla, category and store strings for Amazon-VG — so those players measure a defensible notion of item similarity, but not the curated metadata MovieLens provides. Second, the duplicate-injection diagnostics of Section 9 were run before the replacement and are unaffected by it, since exchangeability of a clone does not depend on the semantics of the cloned source.

The full suite is uneven across corpora

Robustness sweeps (Figure 5), segment analysis, and fusion are strongest on MovieLens-1M, the only corpus that clears the candidate-recall gate. Gowalla required a candidate pool of 11 623 items and still reaches only 0.450 recall (Section 8); its attribution is the least comparable of the three, and we report it for the relative LOO-versus-Shapley contrast only.

Table 7 Observed cost of retiring each source on MovieLens-1M, measured end-to-end under the reported protocol. Negative loss means removal *improved* the metric. The Shapley column is recomputed inside this experiment and is offset from Table 6 by a constant +0.00038 on every source, because the retirement game draws its own baseline permutation; see the note on baseline sensitivity in Section 4. Rankings, which are what the comparison uses, are unaffected. Amazon-VG reproduces the pattern ($\tau = 1.00$ for LOO, 0.60 for Shapley); its table is in the artefact

Source	Observed loss	Recall after	Shapley	LOO
<i>cf</i>	−0.00531	0.690	+0.01564	−0.00501
<i>rec</i>	−0.00062	0.782	+0.00088	+0.00044
<i>ct</i>	−0.00006	0.785	+0.00024	+0.00083
<i>pop</i>	+0.00262	0.723	+0.00680	+0.00098
<i>seq</i>	+0.01636	0.659	+0.02981	+0.01673
Rank agreement with observed loss			$\tau = +0.20$	$\tau = +1.00$

13 Which Estimand, and Does Attribution Predict Retirement Cost?

Two experiments added in revision test whether the paper’s framing survives contact with the decision it motivates.

Three estimands

Our main game refits the fusion head per coalition on a candidate set built from *all* sources. Two alternatives isolate what that choice contributes: v_{fixed} holds the deployed grand-coalition head fixed and merely masks it to S ; v_{e2e} lets each coalition retrieve its own candidates and refit, so a removed source loses its retrieval role too. On MovieLens-1M all three rank *seq* first and agree closely (Kendall $\tau = 1.00$ between refit and fixed heads, 0.80 between either and end-to-end); on Amazon-VG all three rank *cf* first, with $\tau = 0.80$ between the refit head and each alternative. The ranking-stage restriction is therefore consequential in principle but does not change the top-ranked source on either corpus — a scope limitation we can now quantify rather than merely declare.

Retirement simulation, and a negative result

The stronger test is whether attribution predicts what removing a source actually costs. We removed each source *entirely* — from retrieval and fusion — rebuilt candidates from the survivors, refitted, and measured the true end-to-end NDCG@10 loss.

Which LOO is being tested

The LOO column of Table 7 is LOO_{rank} : it comes from the same fixed-candidate game as the Shapley column, so the comparison is like for like. It is *not* LOO_{e2e} , which would equal the observed loss by construction (Equation 1) and could not be evaluated as a predictor of it. The result below is therefore not the tautology it might appear to

Table 8 Retirement-rank agreement over ten seeds. LOO is not exactly 1.00 on every seed — reporting it as such from a single run would have overstated it — but it is close on all three corpora and beats Shapley on each. The last two columns give the fraction of seeds on which each rule identifies the observed cheapest source to retire under this protocol

Corpus	τ LOO (95% CI)	τ Shapley (95% CI)	LOO correct	Shapley correct
MovieLens-1M	0.96 [0.91, 1.00]	0.20 [0.20, 0.20]	100%	0%
Amazon-VG	0.94 [0.88, 1.00]	0.76 [0.68, 0.84]	80%	50%
Gowalla	1.00 [1.00, 1.00]	0.80 [0.80, 0.80]	100%	0%

be. It says something more useful: a quantity computable *without* re-running retrieval predicts the expensive end-to-end outcome well, and predicts it far better than the Shapley value of the same game.

LOO_{rank} **predicts retirement cost better than Shapley** (Table 7). On seed 42 its ranking matches the observed loss exactly ($\tau = 1.00$ against 0.20; exact two-sided permutation $p = 0.017$ and $p = 0.82$ at $n = 5$ sources), and it identifies *cf* as the cheapest source to retire where Shapley selects *ct*. We write *observed* rather than *true* loss throughout: this is a measured realisation under the reported protocol, not a population quantity.

The result holds across seeds and across all three corpora

An earlier version reported this from single runs on two corpora. Over ten seeds (Table 8) the picture is both broader and slightly more measured. LOO’s mean rank agreement is 0.96, 0.94 and 1.00; Shapley’s is 0.20, 0.76 and 0.80. LOO is therefore *not* perfect on every seed, and we correct the earlier claim that it was. It nevertheless dominates on every corpus, and the practical gap is starker than the correlations suggest: LOO identifies the observed cheapest source to retire on 100%, 80% and 100% of seeds, against 0%, 50% and 0% for Shapley. On two of three corpora Shapley never once picks the right source to switch off.

The result is also consistent with the theory rather than a contradiction of it. Removing *cf* *improves* end-to-end NDCG@10 by 0.00531 while dropping candidate recall from 0.748 to 0.690: the collaborative signal is substantially redundant with *seq* and *pop* — the pairwise interaction indices are -0.056 for *cf|seq* and -0.026 for *cf|pop*, the two most negative pairs — so in the grand coalition its scores add noise the others already cover. LOO measures exactly that grand-coalition quantity, which is precisely the quantity a single-source retirement decision needs. Shapley averages over coalitions in which *cf* has no substitute, and correctly reports that the capability has value; but that is not the question a retirement decision asks.

The two methods therefore answer different questions, and our earlier framing conflated them. LOO is the right instrument for “what happens if I remove this one source and keep the rest?”. Shapley is the right instrument for “how should credit for the system’s quality be divided among sources?”, which is the question that arises when allocating engineering effort across a roadmap rather than deleting a component today. We now claim only the latter.

Table 9 Which quantity supports which decision, on the evidence in this paper. The last row is the one we cannot yet fill: allocating future effort needs cost and improvement potential, neither of which any attribution supplies

Decision	Appropriate quantity	Evidence
Remove one deployed source	LOO _{e2e} (by definition); LOO _{rank} is an effective proxy	$\tau = 0.94\text{--}1.00$, 10 seeds (Section 13)
Divide credit for current quality	Shapley or a declared semivalue	efficiency, symmetry (Section 9)
Locate redundancy between sources	pairwise interaction index	Table 5
Decide where to invest next	<i>not established</i>	requires cost and headroom

14 Discussion

Which quantity answers which question

The paper’s central practical result is that source allocation and source removal are different estimands, and that using one for the other’s question gives the wrong answer. Table 9 states the mapping we can defend from the evidence here.

What a positive Shapley value does and does not license

When SignalShap assigns collaborative filtering +0.0153 on MovieLens while ablation assigns -0.0050 , the correct reading is not that ablation is mistaken. It is that under the declared ranking-stage game, *cf* contributes in the coalitions where its substitutes are absent, and that removing it from the deployed system nevertheless improves the metric because those substitutes are present. Both statements are true of the same system. An allocation is a statement about a game; an intervention is a statement about a deployment.

The limits of the allocation reading

We motivate allocation by analogy with dividing engineering effort, and we should be honest that this analogy is untested. A source with high current allocated value is not necessarily the source where additional work pays best: that depends on marginal improvement potential, engineering cost, latency, and maintenance burden, none of which enter v . We therefore present the allocation as a description of the current system, not as a prescription for the roadmap, and we regard closing that gap — an attribution whose payoff includes cost — as the most useful next step for this line of work.

On the strength of the retirement result

LOO ranks retirement cost better than Shapley on all three corpora and over ten seeds. With five sources the exact two-sided permutation p for $\tau = 1.00$ is 0.017, so a single corpus and a single seed would be suggestive rather than conclusive; three corpora and ten seeds make it substantially stronger. The decision-level figures are more striking than the correlations: on MovieLens and Gowalla, Shapley never once selects the observed cheapest source to retire. It remains a result about five-source hybrids under one deployment protocol, and we would not extrapolate it to systems

with many more components, where the grand coalition is a less special point in the lattice.

15 Threats to Validity

Construct validity: what the game measures

The candidate set is the union of *all* sources’ top- N lists and is held fixed across coalitions. This is deliberate — it makes $v(S)$ comparable, since a coalition-specific pool would hand the grand coalition a retrieval advantage — but it has a consequence we did not previously state clearly. Every coalition, including the empty one, ranks items that absent sources helped retrieve. SignalShap therefore estimates *ranking-stage* credit, not end-to-end subsystem value. Retiring a source in production would also remove its retrieved candidates, an effect this game excludes by construction. Claims about subsystem investment should be read with that scope in mind, and separating v_{rank} from a coalition-retrieval v_{e2e} is the most valuable extension we can identify.

Construct validity: what the validation establishes

The duplicate experiment verifies symmetry and efficiency on real fitted games. It does not establish absolute correctness of the magnitudes, and no experiment here does, because no independent ground-truth attribution exists for these systems. Analytic games with a fully known Shapley vector complement this and are reported in Section 10.1: all six reproduce exactly, including the null-player and harmful cases that the symmetry diagnostic cannot reach.

Statistical conclusion validity

All users share the same fitted source models, so treating users as independent understates dependence. The fusion comparison against the globally tuned head is not merely underpowered: at $d_z = -0.003$ on MovieLens the user-level Wilcoxon test is the wrong instrument, because a large n can render a practically empty difference significant. We therefore pre-specified a smallest meaningful difference of 0.002 NDCG@10 and report a two-level bootstrap over seeds and then users, together with a TOST equivalence test (Appendix A). The conclusion is equivalence, not superiority.

Two limitations of that analysis remain. The main study uses three seeds and the revision analyses ten; ten outer clusters is adequate for a bootstrap but not generous, and we report intervals rather than point estimates throughout for that reason. And the hierarchical treatment is applied to the fusion-versus-global comparison, which is the one where the difference is small enough for dependence to matter; the comparisons against LightGCN, SASRec, uniform, and popularity are separated by one to two orders of magnitude and are reported with user-level tests, which we regard as sufficient for differences of that size but not as model-level inference.

Amazon-VG is the one case where fusion beats the global head with a corrected p below 0.05 ($p = 0.019$). We do not present this as a positive result: $d_z = 0.014$, and 6 951 of 7 120 users are exact ties. It is a statistically detectable difference with no practical content, and reporting it otherwise would contradict the equivalence conclusion the same data support.

Internal validity

The coalition heads and the coalition values are fitted and evaluated on *disjoint* folds: the ridge head for each S is fitted against validation targets, and $v(S)$ is the NDCG of the resulting ranking against held-out test items. The main attribution result is therefore not in-sample, and a continuous-integration check asserts the separation so a refactor cannot reintroduce it. The attribution-derived fusion weights of Appendix A are a different matter: there the same validation fold supplies both the coalition heads and the attributions used to build the weights, which is an in-sample optimisation bias even though no test label is touched. Proper cross-fitting would be required to make that a positive claim. Since the measured outcome is equivalence with a simpler baseline, the bias works *against* the conclusion we draw, and we report the experiment as a negative result rather than repairing it. Baseline tuning budgets for LightGCN and SASRec were modest and are not matched to the effort spent on our own pipeline; the absolute figures for those models should be read as reference points under our protocol rather than as their attainable best.

External validity

One subsample per corpus, drawn to fit a memory budget, means cross-corpus results may be sample-specific. The three corpora vary in density, catalogue size, metadata provenance, and sampling simultaneously, so this is not a controlled density comparison; the density ordering is clean — 0.075%, 0.469%, 2.697%, a $36\times$ spread with no adjacent pair closer than $5.7\times$ — but the corpora are not otherwise matched, and we do not attribute the differences between them to density alone.

16 Conclusion

We formulated hybrid-recommender source attribution as a five-player cooperative game and computed Shapley values by exact enumeration over all 32 coalitions, avoiding the estimator variance that forces feature-level Shapley to sample. We showed that a natural statement of the redundancy property is false without monotonicity, gave the counterexample, and audited the hypothesis rather than assuming it.

The strongest evidence is the set of invariant checks. Injecting a duplicate into a real system creates a constraint that holds independently of the data — exchangeable players must receive equal credit — and SignalShap satisfies it to numerical precision, alongside efficiency to 6.9×10^{-18} across all three corpora. Attribution values are otherwise difficult to falsify by inspection, so establishing that the implementation realises its axioms on real fitted games is a meaningful, if bounded, form of validation. Under the same condition, ablation assigns zero credit to both substitutes.

We are explicit about what remains open, and about one question we answered against ourselves. These checks do not identify the absolute correct credit, and the game measures ranking-stage rather than end-to-end contribution. The decisive experiment — whether Shapley credit leads to better subsystem decisions than ablation — we did run, as an end-to-end retirement simulation, and ablation won it on all three corpora (Section 13). We therefore claim credit allocation, not retirement guidance. Two corpora also fall below the candidate-recall gate, and on Gowalla lifting it would

require a candidate pool approaching the whole catalogue, at which point the fixed-candidate game no longer models the two-stage architecture it was built for. That is a cost boundary on the design, not an impossibility, and we state it as such after withdrawing a stronger claim made in an earlier version.

Future work

Four directions follow directly from what we could not settle. First, a cost-aware estimand: allocation describes the current system, but deciding where to invest needs marginal improvement potential and engineering cost inside the payoff, and neither is in v . Second, scaling beyond a handful of players, where exhaustive enumeration fails and the exactness that motivates this design must be traded against sampling error — a comparison of exact and sampled values across n would establish where the boundary lies. Third, corpora dense enough for fixed-candidate retrieval to clear the recall gate, since only one of ours does. Fourth, richer players: graph-neural and Transformer components, privacy-bearing signals, and cascade rather than parallel hybrids, all of which change the retrieval structure the game assumes.

A Attribution-Derived Fusion: A Negative Result

This appendix reports an experiment that did not work. We include it because the natural next step after allocating credit is to *use* that allocation, and a reader who asks whether attribution-derived weights help deserves the measured answer rather than silence. They do not.

Two questions separate here: whether attributions *differ* across user segments, and whether exploiting that difference *pays*.

Segments and the weight map

An earlier version of this paper left both definitions implicit, which made the applied claim unreproducible. Segments are a 2×2 split on training behaviour only, computed before any test data is touched: activity (above or below the median interaction count) crossed with recency skew, the fraction of a user’s interactions falling in the newer half of that user’s own timespan, again split at the median. This yields four segments; users with no training interactions fall in the first.

Segment attributions are mapped to fusion weights by a softmax with shrinkage toward the global profile. For segment s and source g ,

$$q_{s,g} = \frac{\exp(\varphi_{s,g}/\tau)}{\sum_{h \in \mathcal{G}} \exp(\varphi_{s,h}/\tau)}, \quad a_{s,g} = (1 - \rho) q_{s,g} + \rho q_{\text{global},g}. \quad (11)$$

The softmax is not cosmetic. Our games are non-monotone, so $\varphi_{s,g}$ can be negative, and using raw values as weights would *invert* a source’s scores rather than down-weight it — a qualitatively different operation from the one the attribution licenses. Shrinking a noisy per-segment estimate toward a pooled one is the classical variance-reduction trade [19]. The temperature τ and shrinkage ρ are selected on the validation fold and frozen before any test evaluation.

Heterogeneity is real, but only on the dense corpus

On MovieLens-1M, 8 of 30 segment-pair contrasts are significant at $\alpha = 0.05$ uncorrected, against 1.5 expected by chance. On the two sparse corpora the counts are 4 of 30 (Gowalla) and 5 of 30 (Amazon-VG). These are above the chance rate but well below MovieLens, and after Holm correction within each corpus none of the three corpora retains a significant contrast — a fact we state because the uncorrected counts alone would overstate the evidence. We therefore claim suggestive segment heterogeneity in the dense setting and no established heterogeneity on sparse data, and note that this is a plausible reason the closed loop does not deliver. Figure 6 plots the per-segment profiles.

A multi-comparison test, and why it must be read narrowly

A Wilcoxon test per corpus establishes a difference on that corpus, not a ranking across corpora, so we add a Friedman omnibus over the six methods blocked by corpus, with a Nemenyi post-hoc. The omnibus is significant ($\chi^2 = 14.24$, $p = 0.014$) and the mean ranks order the methods as one would expect — SignalShap-Fuse 1.33, globally tuned head 1.67, uniform 3.00, LightGCN 4.00, popularity 5.33, SASRec 5.67 — but with three blocks the critical difference is 4.35 and *no pair is separable*.

Two caveats, the second of which we flag against our own analysis. First, three blocks is too few; adding corpora, not seeds, is what would fix it. Second, and more seriously, this test pools absolute NDCG@10 across all three corpora, including the two that Section 8 excludes from absolute comparison against the neural references because they fail the recall gate. That is inconsistent with our own stated policy. We keep the test only because its conclusion is negative — nothing separates — so it cannot manufacture a claim the gate was meant to prevent; a positive result here would have had to be discarded. Readers should treat the mean ranks as descriptive and the per-corpus tests of Section 11 as the reportable statistics.

Fusion weights never see test outcomes

Because the characteristic function of Section 4 is evaluated on test interactions, deriving fusion weights from those same attributions would be circular. We avoid this: the fusion heads, the segment-specific weights, the choice of head family, and the shrinkage strength are all fitted on the *validation* fold only, and are frozen before a single test evaluation. The Shapley values reported in Section 11 are diagnostic output; they are not inputs to the weights. This separation is enforced in code and covered by the released test suite.

Attribution-derived fusion beats weak comparators and nothing else

Weights derived from the segment-level attributions outperform every *independent* comparator at Holm-corrected $p < 0.001$: uniform fusion (+0.0120, $d_z = 0.098$), a popularity reference (+0.0450, $d_z = 0.231$), LightGCN [14] (+0.0293, $d_z = 0.156$), and SASRec [21] (+0.0462, $d_z = 0.243$), at NDCG@10 = 0.06327 (Figure 7). We attach little weight to these: the neural references received smaller tuning budgets, and every effect size is below 0.25.

Against a globally tuned head of the same family ($\text{NDCG@10} = 0.06324$) the difference is $+0.00003$ ($d_z = 0.001$, Holm-corrected $p = 0.77$) — indistinguishable from zero. The sign of this comparison has flipped between revisions while its magnitude stayed within noise, which is itself the clearest evidence that no effect is resolvable here. Under the per-user normalisation of Eq. (5) the sign of this comparison changed while its magnitude stayed within noise, which is itself the clearest evidence that no effect is resolvable here. **We therefore make no claim of any advantage over a well-tuned global head**, and note that the comparisons against LightGCN and SASRec are conditional on tuning budgets that were not matched to our own pipeline.

We now test this directly rather than describing it as underpowered. Because users share the same fitted source models and coalition heads, a user-level test conditions on one draw of the training procedure; we therefore use a two-level bootstrap that resamples seeds and then users within each seed. It gives a mean difference of $+0.00005$ with a 95% interval of $[-0.00022, +0.00032]$, which spans zero.

We also pre-specified a smallest meaningful difference of 0.002 NDCG@10 and ran a two one-sided test. It rejects a true difference larger than that margin ($p < 0.001$), so the two configurations are **statistically equivalent for practical purposes** — a positive claim, and a stronger one than the absence of significance. The rank-biserial correlation is $+0.007$. We therefore make the claim the evidence supports: SignalShap attribution yields fusion weights no better than a tuned global head, and its advantage over a carefully tuned global head is not established at this scale.

The contribution stands on the validation of Section 9 and the attribution results of Section 11. Fusion demonstrates that the attributions are actionable; it is not the evidence that they are correct.

Supplementary information. Code, configuration, cached score matrices, and every artefact JSON backing the numbers reported here are available at <https://github.com/mouadloughichi/signalshap-code>. The repository contains the frozen configuration, one command per table and figure, and a test suite covering the efficiency, symmetry, and monotonicity invariants. An immutable archived release with a DOI will be deposited at acceptance.

Acknowledgements. Not applicable.

Declarations

- **Funding.** The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.
- **Competing interests.** The authors declare none.
- **Ethics approval.** Not applicable: no human-subjects data or biological material. Public benchmarks are used under the licenses named in the artefact.
- **Data availability.** MovieLens-1M is used under the GroupLens research-use license; Gowalla check-ins from the SNAP collection; and the Video Games category of the Amazon Reviews 2023 corpus. All are public. Code and artefacts will be archived with a DOI at publication.
- **Code availability.** Public repository, MIT license.

- **Author contributions.** **M. Louhichi:** conceptualisation, methodology, software, formal analysis, investigation, writing — original draft, visualisation. **R. Nesmaoui:** software, validation, data curation, writing — review & editing. **M. Lazaar:** supervision, methodology, resources, writing — review & editing.
- **Use of AI tools.** Large language models were used for code scaffolding, editorial review, and consistency checking. All mathematical claims were independently verified, and the counterexample in Section 6 is machine-checked in the artefact’s test suite. The authors take full responsibility for the content.

References

- [1] Burges C, Shaked T, Renshaw E, et al (2005) Learning to rank using gradient descent. In: Proceedings of the 22nd International Conference on Machine Learning (ICML), pp 89–96, <https://doi.org/10.1145/1102351.1102363>
- [2] Burke R (2002) Hybrid recommender systems: Survey and experiments. *User Modeling and User-Adapted Interaction* 12(4):331–370. <https://doi.org/10.1023/A:1021240730564>
- [3] Chang H, Lee S, Choi J (2025) Rethinking shapley value for negative interactions in non-convex games. In: International Conference on Learning Representations (ICLR)
- [4] Cho E, Myers SA, Leskovec J (2011) Friendship and mobility: User movement in location-based social networks. In: Proceedings of the 17th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp 1082–1090, <https://doi.org/10.1145/2020408.2020579>
- [5] Chowdhury T, Zick Y, Allan J (2025) Rankshap: Shapley value based feature attributions for learning to rank. In: International Conference on Learning Representations (ICLR)
- [6] Covert I, Lundberg SM, Lee SI (2021) Explaining by removing: A unified framework for model explanation. *Journal of Machine Learning Research* 22(209):1–90
- [7] Fumagalli F, Muschalik M, Kolpaczki P, et al (2024) Kernelshap-iq: Weighted least square optimization for shapley interactions. In: Proceedings of the 41st International Conference on Machine Learning (ICML)
- [8] Ghorbani A, Zou J (2019) Data shapley: Equitable valuation of data for machine learning. In: Proceedings of the 36th International Conference on Machine Learning (ICML), pp 2242–2251
- [9] Ghorbani A, Zou J (2020) Neuron shapley: Discovering the responsible neurons. In: Advances in Neural Information Processing Systems (NeurIPS)

- [10] Grabisch M, Roubens M (1999) An axiomatic approach to the concept of interaction among players in cooperative games. *International Journal of Game Theory* 28(4):547–565. <https://doi.org/10.1007/s001820050125>
- [11] Han D, Wooldridge M, Rogers A, et al (2022) Replication-robust payoff-allocation for machine learning data markets. In: *IEEE Transactions on Artificial Intelligence*, <https://doi.org/10.1109/TAI.2022.3212804>
- [12] Harper FM, Konstan JA (2015) The movielens datasets: History and context. *ACM Transactions on Interactive Intelligent Systems* 5(4):1–19. <https://doi.org/10.1145/2827872>
- [13] He B, Ma C (2024) Interpretable triplet importance for personalized ranking. In: *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM)*, <https://doi.org/10.48550/arXiv.2407.19469>
- [14] He X, Deng K, Wang X, et al (2020) Lightgcn: Simplifying and powering graph convolution network for recommendation. In: *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp 639–648, <https://doi.org/10.1145/3397271.3401063>
- [15] Heuss M, de Rijke M, Anand A (2025) Rankingshap: Listwise feature attribution explanations for ranking models. In: *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, <https://doi.org/10.48550/arXiv.2403.16085>
- [16] Holm S (1979) A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics* 6(2):65–70
- [17] Hou Y, Li J, He Z, et al (2024) Bridging language and items for retrieval and recommendation. In: *arXiv preprint arXiv:2403.03952*, <https://doi.org/10.48550/arXiv.2403.03952>
- [18] Hu Y, Koren Y, Volinsky C (2008) Collaborative filtering for implicit feedback datasets. In: *Proceedings of the 8th IEEE International Conference on Data Mining (ICDM)*, pp 263–272, <https://doi.org/10.1109/ICDM.2008.22>
- [19] James W, Stein C (1961) Estimation with quadratic loss. *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability* 1:361–379
- [20] Järvelin K, Kekäläinen J (2002) Cumulated gain-based evaluation of ir techniques. *ACM Transactions on Information Systems* 20(4):422–446. <https://doi.org/10.1145/582415.582418>
- [21] Kang WC, McAuley J (2018) Self-attentive sequential recommendation. In: *Proceedings of the 18th IEEE International Conference on Data Mining (ICDM)*, pp

197–206, <https://doi.org/10.1109/ICDM.2018.00035>

- [22] Karczmarz A, Michalak T, Mukherjee A, et al (2022) Improved feature importance computation for tree models based on the banzhaf value. In: Proceedings of the 38th Conference on Uncertainty in Artificial Intelligence (UAI)
- [23] Levy O, Goldberg Y (2014) Neural word embedding as implicit matrix factorization. In: Advances in Neural Information Processing Systems (NeurIPS), pp 2177–2185
- [24] Louhichi M, Nesmaoui R, Lazaar M (2025) Game theory meets explainable AI: An enhanced approach to understanding black box models through shapley values. *International Journal of Advanced Computer Science and Applications* 16(7). <https://doi.org/10.14569/IJACSA.2025.0160770>
- [25] Lundberg SM, Lee SI (2017) A unified approach to interpreting model predictions. In: Advances in Neural Information Processing Systems (NeurIPS), pp 4765–4774
- [26] Lundberg SM, Erion G, Chen H, et al (2020) From local explanations to global understanding with explainable ai for trees. *Nature Machine Intelligence* 2(1):56–67. <https://doi.org/10.1038/s42256-019-0138-9>
- [27] Ni J, Li J, McAuley J (2019) Justifying recommendations using distantly-labeled reviews and fine-grained aspects. In: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP), pp 188–197, <https://doi.org/10.18653/v1/D19-1018>
- [28] Pliatsika V, Stoyanovich J, et al (2025) Sharp: Explaining rankings and preferences with shapley values. *Proceedings of the VLDB Endowment* 18:4131–4144
- [29] Ribeiro MT, Singh S, Guestrin C (2016) “why should i trust you?”: Explaining the predictions of any classifier. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp 1135–1144, <https://doi.org/10.1145/2939672.2939778>
- [30] Rozemberczki B, Sarkar R (2021) The shapley value of classifiers in ensemble games. In: Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM), pp 1558–1567, <https://doi.org/10.1145/3459637.3482302>
- [31] Rozemberczki B, Watson L, Bayer P, et al (2022) The shapley value in machine learning pp 5572–5579. <https://doi.org/10.24963/ijcai.2022/778>
- [32] Shah H, Ilyas A, Madry A (2024) Decomposing and editing predictions by modeling model computation. In: Proceedings of the 41st International Conference on Machine Learning (ICML)

- [33] Shapley LS (1953) A value for n -person games. In: Kuhn HW, Tucker AW (eds) Contributions to the Theory of Games, Volume II, Annals of Mathematics Studies, vol 28. Princeton University Press, p 307–317, <https://doi.org/10.1515/9781400881970-018>
- [34] Sundararajan M, Najmi A (2020) The many shapley values for model explanation. In: Proceedings of the 37th International Conference on Machine Learning (ICML), pp 9269–9278
- [35] Sundararajan M, Taly A, Yan Q (2017) Axiomatic attribution for deep networks. In: Proceedings of the 34th International Conference on Machine Learning (ICML), pp 3319–3328
- [36] Sundararajan M, Dhamdhere K, Agarwal A (2020) The shapley taylor interaction index. In: Proceedings of the 37th International Conference on Machine Learning (ICML), pp 9259–9268
- [37] Tsai CP, Yeh CK, Ravikumar P (2023) Faith-shap: The faithful shapley interaction index. In: Journal of Machine Learning Research, pp 1–42
- [38] Wang T, Rausch J, Zhang C, et al (2020) A principled approach to data valuation for federated learning. In: Federated Learning: Privacy and Incentive. Springer, pp 153–167, https://doi.org/10.1007/978-3-030-63076-8_11
- [39] Wilcoxon F (1945) Individual comparisons by ranking methods. Biometrics Bulletin 1(6):80–83. <https://doi.org/10.2307/3001968>
- [40] Zangerle E, Bauer C (2022) Evaluating recommender systems: Survey and framework. ACM Computing Surveys 55(8):1–38. <https://doi.org/10.1145/3556536>
- [41] Zhang H, Singh H, Ghassemi M, et al (2023) ”why did the model fail?”: Attributing model performance changes to distribution shifts. In: Proceedings of the 40th International Conference on Machine Learning (ICML)
- [42] Zhang Y, Chen X (2020) Explainable recommendation: A survey and new perspectives. Foundations and Trends in Information Retrieval 14(1):1–101. <https://doi.org/10.1561/15000000066>

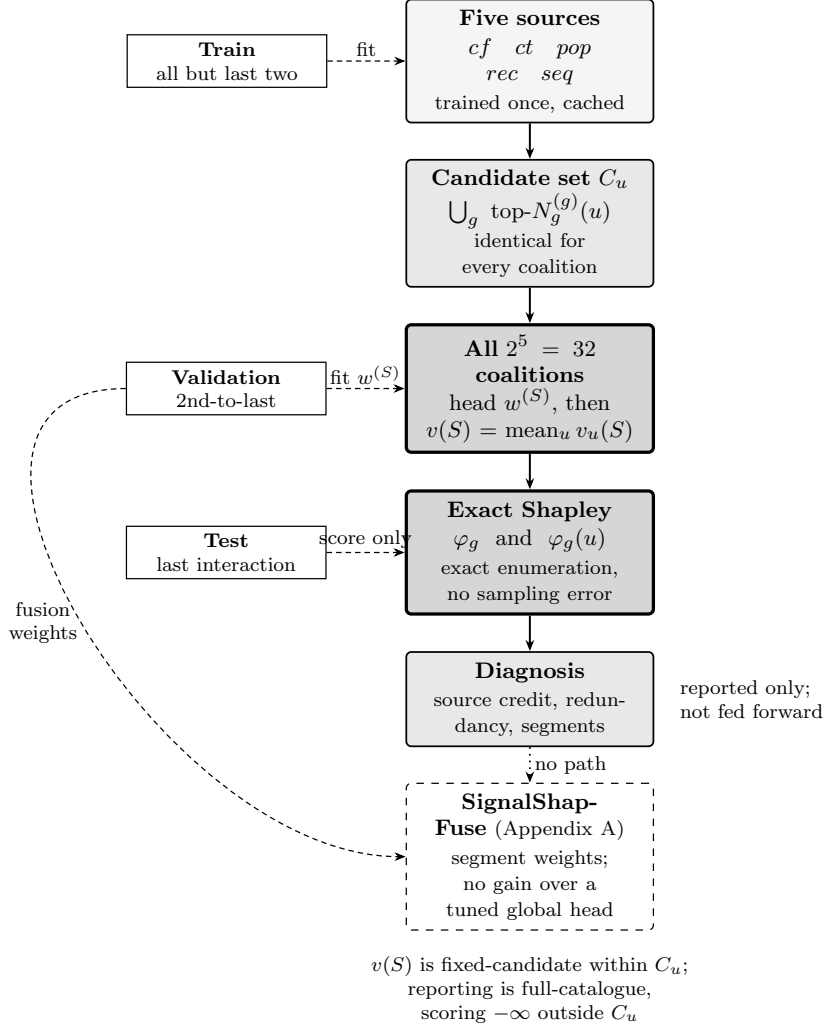


Fig. 1 The SignalShap pipeline, with data folds made explicit. Sources are fitted on train; every coalition head $w^{(S)}$ and every fusion weight is fitted on *validation*; test interactions are used only to score. The candidate set C_u is the union of all sources’ top- N lists and is identical for every coalition, so differences in $v(S)$ reflect fusion within S rather than retrieval from a pool that favours it. Because $v(S)$ is evaluated on test interactions, the Shapley output is reported as diagnosis and is deliberately *not* an input to the fusion weights (dotted arrow); that separation is what keeps the downstream comparison of Appendix A non-circular

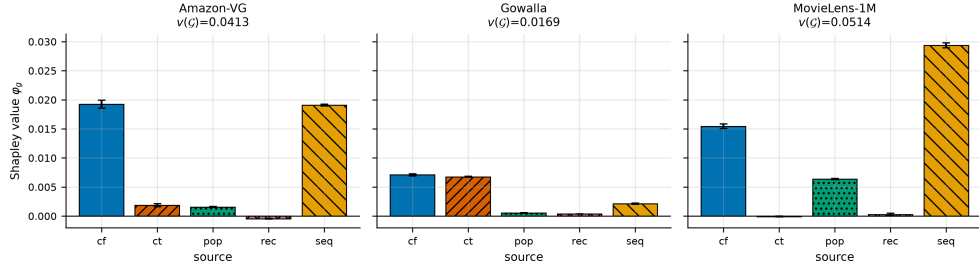


Fig. 2 Per-source Shapley values with error bars spanning ± 1.96 standard errors over the three main-study seeds. Intervals appear in the main text because v is fitted: exactness is a property of the aggregation, not of the attribution end to end.

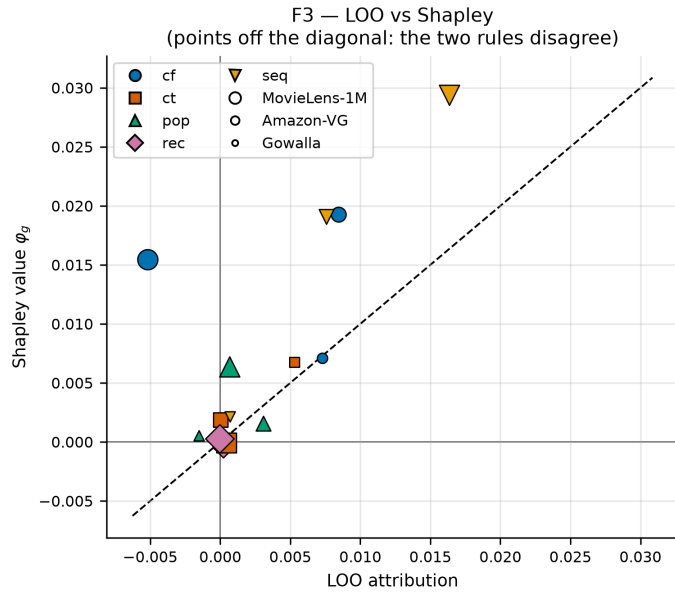


Fig. 3 SignalShap attribution against leave-one-out ablation. Points on the diagonal are cases where the two agree; the upper-left quadrant holds sources SignalShap credits as valuable that ablation scores below zero.

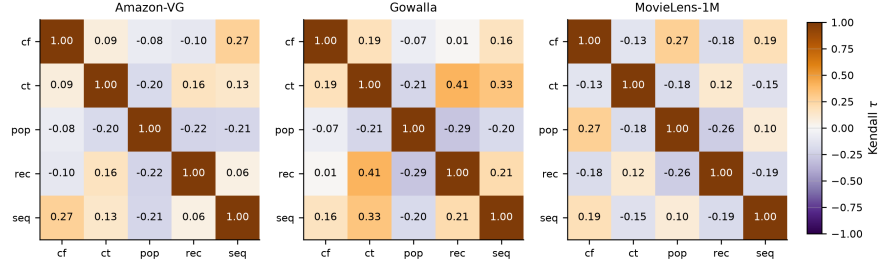


Fig. 4 Redundancy between sources, measured as Kendall τ between per-user score vectors. The pop - cf and rec - ct pairs are expected to overlap by construction, as stated in advance in Section 7.

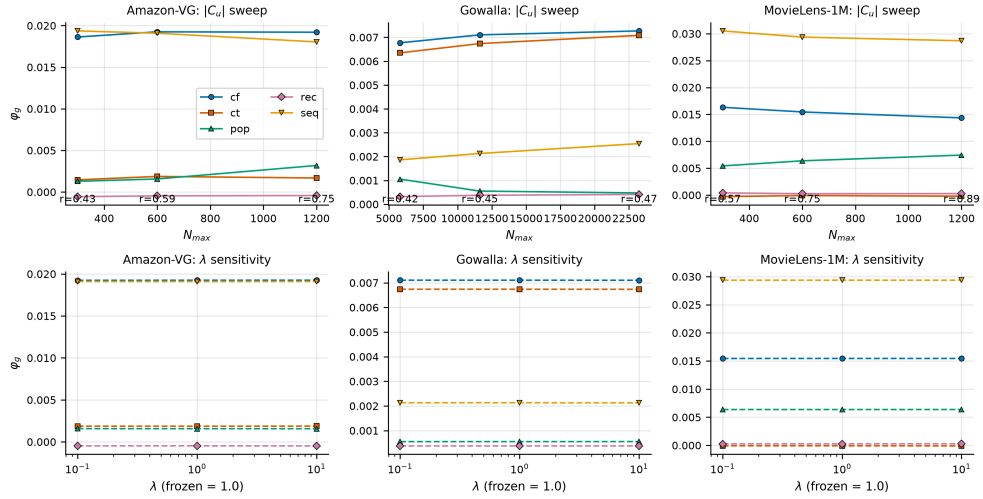


Fig. 5 Robustness. Candidate recall is annotated per cell of the $|C_u|$ sweep, because changing $|C_u|$ moves the recall ceiling and hence the level of v ; without it the cells are not comparable. The λ panel is sensitivity only and is never used for selection.

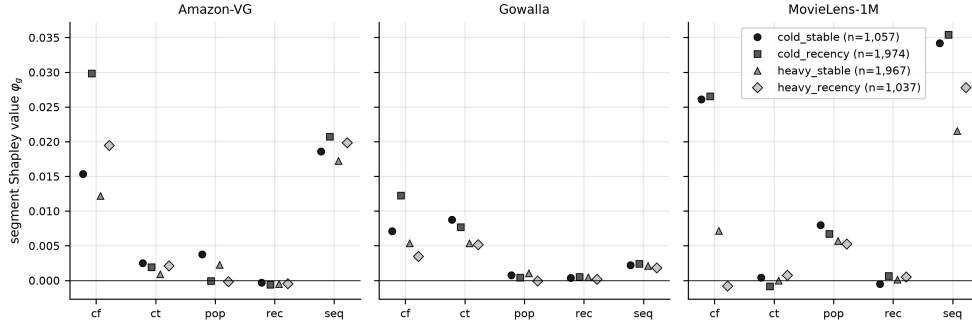


Fig. 6 Segment-level Shapley values, with segment sizes in the legend. Overlapping markers within a source indicate little heterogeneity to exploit. We use a shared linear axis rather than the radar chart of an earlier version: radar encodes magnitude as radius and therefore area, which exaggerates differences quadratically and has no natural zero, and our values can be negative.

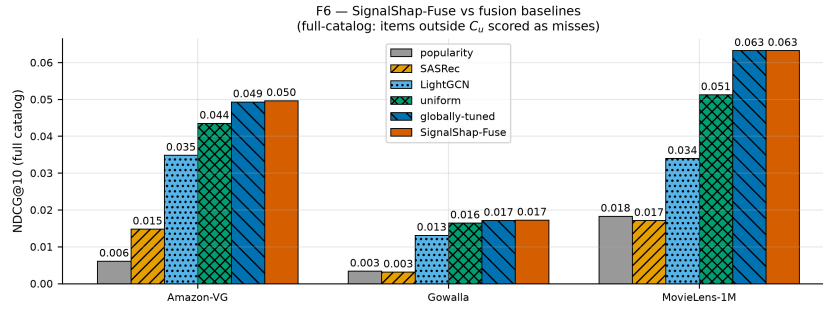


Fig. 7 Fusion comparison under the full-catalogue protocol, showing every comparator named in the text. SignalShap-Fuse and the globally tuned head are visually indistinguishable on all three corpora, which is the point of this appendix. The neural and heuristic references are lower, but they received smaller tuning budgets and two of the three corpora fail the recall gate, so no ranking claim is made against them.